Real Analysis

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Introduction

Real analysis provides the rigorous foundation for the study of limits, continuity, convergence, measure, and integration. These notes develop the subject from its basic set-theoretic and topological foundations to the theory of measure and the Lebesgue integral, and then proceed to signed measures, differentiation, function spaces, and product measures.

The first part introduces set theory, cardinality, metric spaces, compactness, approximation, and density. These topics provide the language and structural tools needed throughout the course. In particular, completeness and compactness play a central role in understanding convergence, while the Baire Category Theorem and the Stone–Weierstrass Theorem illustrate two fundamental principles concerning the structure and approximation of functions.

The second part develops measure theory, beginning with rings, algebras, semi-rings, outer measures, and the construction of Lebesgue measure. Measurable functions and different modes of convergence are then studied, including almost everywhere convergence, convergence in measure, Egorov's Theorem, and Lusin's Theorem. Based on this framework, the Lebesgue integral is constructed first for simple functions and non-negative functions and then for general integrable functions. The principal convergence results—the Monotone Convergence Theorem, Fatou's Lemma, and the Dominated Convergence Theorem—form the core tools for exchanging limits and integrals.

The later parts extend the theory to signed measures and the Radon–Nikodym Theorem, develop the connections between differentiation and integration through functions of bounded variation and absolutely continuous functions, and introduce the L^p spaces together with their fundamental inequalities, completeness, approximation properties, and duality. Finally, product measures and the Fubini–Tonelli theorems provide a rigorous framework for integration on product spaces.

In addition to the original lecture material, these notes include supplementary explanations, detailed proofs, and remarks intended to clarify arguments that are abbreviated in the original notes and to make explicit the connections between important definitions and theorems.

Chapter 1

Elements of Set Theory

1 Elements of Set Theory

1.1 Naive Set Theory and Paradoxes

In naive set theory, we intuitively understand a set as a collection of distinct objects, considered as an object in its own right. However, relying solely on natural descriptions to define sets can lead to contradictions.

Example 1.1 (The Barber Paradox)**.** Consider a town with a barber who shaves all those and only those who do not shave themselves.

- If the barber shaves himself, he belongs to the group of people he does not shave. Contradiction.
- If the barber does not shave himself, he belongs to the group of people he must shave. Contradiction.

This implies such a set of people cannot exist.

Example 1.2 (Russell's Paradox)**.** Let A be the set of all sets that do not contain themselves as elements:

$$A = \{S \mid S \notin S\}.$$

Question: Is $A \in A$?

- If $A \in A$, then by the definition of A, $A \notin A$.
- If $A \notin A$, then by the definition of A, $A \in A$.

Observation 1.3 (Moral)**.** Not all collections defined by a "natural" description constitute a set. To avoid these paradoxes, axiomatic systems (like ZFC) are required, though in this course we will proceed with standard set-theoretic operations while being mindful of these limitations.

Set Operations

Let A, B be sets. We define:

1. **Union:** $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.
2. **Intersection:** $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.
3. **Difference:** $A \setminus B = \{x \mid x \in A \text{ and } x \notin B\}$.
4. **Symmetric Difference:** $A \triangle B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.

Exercise 1.4. Which possible collections of the first 3 operations (union, intersection, difference) generate the symmetric difference?

Proposition 1.5 (Duality Principle / De Morgan's Laws)**.** *Let Ω be a universal set and $A_\alpha \subset \Omega$ for $\alpha \in E$. Then:*

$$\Omega \setminus \left(\bigcup_{\alpha \in E} A_\alpha \right) = \bigcap_{\alpha \in E} (\Omega \setminus A_\alpha).$$

Similarly, the complement of an intersection is the union of complements.

Proof. For any $x \in \Omega$,

$$x \in \Omega \setminus \left(\bigcup_{\alpha \in E} A_\alpha \right) \iff x \notin \bigcup_{\alpha \in E} A_\alpha \iff \forall \alpha \in E,\ x \notin A_\alpha \iff x \in \bigcap_{\alpha \in E} (\Omega \setminus A_\alpha).$$

Hence the two sets are equal. Similarly,

$$\Omega \setminus \left(\bigcap_{\alpha \in E} A_\alpha \right) = \bigcup_{\alpha \in E} (\Omega \setminus A_\alpha).$$

□

1.2 Mappings and Functions

Definition 1.6. Let A, B be sets. A **mapping** (or function) $f : A \to B$ is a correspondence that assigns to every $a \in A$ a unique element $b = f(a) \in B$. Formally, it is a subset of $A \times B$.

Definition 1.7 (Types of Mappings)**.** Let $f : A \to B$.

- **Injective** (one-to-one): $x \neq y \implies f(x) \neq f(y)$.
- **Surjective** (onto): $f(A) = B$, i.e., $\forall b \in B, \exists a \in A$ such that $f(a) = b$.
- **Bijective**: Both injective and surjective. In this case, there exists an inverse mapping $f^{-1} : B \to A$.

1.3 Equivalence Relations and Orderings

1.3.1 Equivalence Relations

Definition 1.8. A relation $\sim$ on a set X (a subset $E \subset X \times X$) is an **equivalence relation** if it satisfies:

1. **Reflexivity:** $x \sim x$ for all $x \in X$.
2. **Symmetry:** $x \sim y \implies y \sim x$.
3. **Transitivity:** $x \sim y$ and $y \sim z \implies x \sim z$.

Theorem 1.9. *An equivalence relation on X partitions X into disjoint equivalence classes:*

$$X = \bigsqcup_{\alpha \in A} X_\alpha,$$

where each $X_\alpha = [x] = \{y \in X \mid y \sim x\}$.

Proof. For every $x \in X$, reflexivity gives $x \sim x$, hence $x \in [x]$. Therefore every point of X belongs to at least one equivalence class.

Now suppose that two equivalence classes $[x]$ and $[y]$ have a common element z. Then $z \sim x$ and $z \sim y$. By symmetry, $x \sim z$, and by transitivity, $x \sim y$. If $w \in [x]$, then $w \sim x$ and $x \sim y$, so $w \sim y$, hence $w \in [y]$. Thus $[x] \subset [y]$. The reverse inclusion follows in the same way, so $[x] = [y]$.

Therefore any two equivalence classes are either equal or disjoint, and their union is X. □

Example 1.10.
1. In $M_n(\mathbb{R})$, similarity of matrices: $A \sim B$ if $A = P^{-1}BP$.
2. In $\mathbb{C}$, $z \sim w$ if $|z| = |w|$ (concentric circles).
3. Cardinality: Sets $A \sim B$ if there exists a bijection between them.

1.3.2 Orderings

Definition 1.11 (Partial Order)**.** A relation $\leq$ on a set X is a **partial ordering** if:

1. **Reflexivity:** $x \leq x$.
2. **Anti-symmetry:** $x \leq y$ and $y \leq x \implies x = y$.

3. **Transitivity:** $x \leq y$ and $y \leq z \implies x \leq z$.

The pair $(X, \leq)$ is called a partially ordered set (poset). Note that not every pair of elements need be comparable.

Definition 1.12 (Total Order and Well-Ordering)**.**

- A set is **totally ordered** if $\forall x, y \in X$, either $x \leq y$ or $y \leq x$.
- A set is **well-ordered** if every non-empty subset $Y \subset X$ has a least element (an element $a \in Y$ such that $a \leq y$ for all $y \in Y$).

Example 1.13.

- $(\mathbb{N}, \leq)$ is well-ordered.
- $(\mathbb{R}, \leq)$ is totally ordered but not well-ordered (e.g., $(0, 1)$ has no minimum).
- 2^A with inclusion $\subseteq$ is a partial order, not total.

1.3.3 Axiom of Choice and Zorn's Lemma

Axiom 1.14 (Axiom of Choice)**.** *For any collection $\{X_\alpha\}_{\alpha \in A}$ of non-empty sets, there exists a set E containing exactly one element from each X_α. Equivalently, the Cartesian product*

$$\prod_{\alpha \in A} X_\alpha$$

is non-empty.

Definition 1.15 (Chain and Upper Bound)**.** Let $(X, \leq)$ be a poset.

- A subset $C \subset X$ is a **chain** if it is totally ordered by $\leq$.
- An element $u \in X$ is an **upper bound** for C if $c \leq u$ for all $c \in C$.
- An element $m \in X$ is **maximal** if $m \leq x \implies m = x$.

Remark 1.16. An upper bound u of a chain C does not have to belong to C.

Lemma 1.17 (Zorn's Lemma)**.** *Let X be a non-empty partially ordered set. If every chain in X has an upper bound in X, then X contains a maximal element.*

Theorem 1.18 (Application: Basis of Vector Space)**.** *Every vector space V has a basis (Hamel basis).*

Sketch. Let $\mathcal{I}$ be the collection of all linearly independent subsets of V, ordered by inclusion. $\mathcal{I}$ is non-empty ($\varnothing \in \mathcal{I}$). If $\mathcal{C}$ is a chain in $\mathcal{I}$, then $\bigcup_{A \in \mathcal{C}} A$ is also linearly independent and serves as an upper bound. By Zorn's Lemma, there exists a maximal independent set B. One can show B spans V.

Additional Details. It remains to justify the two points used in the sketch.

First, let

$$U = \bigcup_{A \in \mathcal{C}} A.$$

To show that U is linearly independent, take finitely many vectors $v_1, \ldots, v_n \in U$. For each i, there exists $A_i \in \mathcal{C}$ such that $v_i \in A_i$. Since $\mathcal{C}$ is a chain under inclusion, the finite collection $A_1, \ldots, A_n$ is totally ordered by inclusion, so one of these sets contains all the others. Hence there exists $A_j \in \mathcal{C}$ containing $v_1, \ldots, v_n$. Since A_j is linearly independent, $v_1, \ldots, v_n$ are linearly independent. Thus $U \in \mathcal{I}$, and clearly $A \subset U$ for every $A \in \mathcal{C}$, so U is an upper bound of $\mathcal{C}$.

By Zorn's Lemma, $\mathcal{I}$ therefore contains a maximal element B. Suppose B does not span V. Then there exists $v \in V \setminus \operatorname{span}(B)$. Hence $B \cup \{v\}$ is still linearly independent, contradicting the maximality of B. Therefore $\operatorname{span}(B) = V$, so B is a Hamel basis of V. $\square$

Chapter 2

Cardinality

2 Cardinality

2.1 Countable and Uncountable Sets

Definition 2.1. We say two sets A and B have the same cardinality (denoted $|A| = |B|$ or $A \sim B$) if there exists a bijection $f : A \to B$.

Definition 2.2.
- A set A is countable if $A \sim \mathbb{N}$ (or is finite).
- A set is at most countable if it is finite or countable.
- A set is uncountable if it is infinite and not countable.

We denote the cardinality of $\mathbb{N}$ by $\aleph_0$.

Theorem 2.3. *Let $\{A_n\}_{n\in\mathbb{N}}$ be a countable collection of at most countable sets. Then*

$$A = \bigcup_{n=1}^{\infty} A_n$$

is at most countable.

Additional Proof. For each n, since A_n is at most countable, its elements can be listed as

$$A_n = \{a_{n1}, a_{n2}, a_{n3}, \ldots\},$$

where finite sets may simply be regarded as lists that terminate. Thus the elements of A may be arranged in an infinite array

$$\begin{array}{cccc} a_{11} & a_{12} & a_{13} & \cdots \\ a_{21} & a_{22} & a_{23} & \cdots \\ a_{31} & a_{32} & a_{33} & \cdots \\ \vdots & \vdots & \vdots & \end{array}$$

and enumerated diagonally:

$$a_{11},\ a_{12},\ a_{21},\ a_{13},\ a_{22},\ a_{31}, \ldots$$

Omitting repeated elements gives an enumeration of A. Hence A is at most countable. □

Proposition 2.4. *Any infinite set contains a countable subset.*

Additional Proof. Let A be infinite. Choose $a_1 \in A$. Since A is infinite, $A \setminus \{a_1\}$ is non-empty, so choose

$$a_2 \in A \setminus \{a_1\}.$$

Inductively, after choosing distinct elements $a_1, \ldots, a_n$, the set

$$A \setminus \{a_1, \ldots, a_n\}$$

is non-empty, so choose a_{n+1} from it. Thus

$$\{a_1, a_2, a_3, \ldots\} \subset A$$

is countable. □

2.2 The Continuum

Definition 2.5. A set E is called a continuum (or has the cardinality of the continuum, denoted $\mathfrak{c}$ or $2^{\aleph_0}$) if $E \sim [0,1]$.

Example 2.6.
- $[a,b] \sim [0,1]$ via linear scaling.
- $(-\pi/2, \pi/2) \sim \mathbb{R}$ via $f(x) = \tan x$.
- $(0,1) \sim [0,1]$. (Proof involves shifting a countable subset).

Additional Detail. Consider the countable subset

$$S = \left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\} \subset (0,1).$$

Define $f : [0,1] \to (0,1)$ by

$$f(0) = \frac{1}{2}, \qquad f(1) = \frac{1}{3}, \qquad f\left(\frac{1}{n}\right) = \frac{1}{n+2} \quad (n \geq 2),$$

and

$$f(x) = x$$

for every other $x \in [0,1]$. This shifts the countable subset

$$0,\ 1,\ \frac{1}{2},\ \frac{1}{3}, \ldots$$

inside $(0,1)$ while leaving all other points fixed. Hence f is a bijection and $(0,1) \sim [0,1]$. □

Theorem 2.7 (Uncountability of the Continuum)**.** *The interval* $[0,1]$ *is not countable.*

Geometric Proof from Lecture. Assume by contradiction that $[0,1]$ is countable, i.e., there exists a bijection $f : \mathbb{N} \to [0,1]$. Let us list the elements:

$$[0,1] = \{x_1, x_2, x_3, \ldots\}.$$

Step 1: Divide $[0,1]$ into 3 equal closed sub-intervals:

$$[0, 1/3], \qquad [1/3, 2/3], \qquad [2/3, 1].$$

Since x_1 is a single point, at least one of these closed intervals does not contain x_1. Choose that interval and call it I_1.
Step 2: Divide I_1 into 3 equal closed sub-intervals. Choose one, call it I_2, such that $x_2 \notin I_2$.
Inductive Step: Having chosen I_{k-1}, divide it into 3 equal parts. Choose one, I_k, such that $x_k \notin I_k$.

We obtain a sequence of nested closed intervals

$$I_1 \supset I_2 \supset \cdots$$

with lengths tending to 0. By the Nested Interval Property of $\mathbb{R}$, the intersection is non-empty:

$$\bigcap_{k=1}^{\infty} I_k = \{\xi\} \neq \varnothing.$$

By construction, for every k, $\xi \in I_k$ and $x_k \notin I_k$, so $\xi \neq x_k$. Thus ξ is a number in $[0,1]$ that is not in the list $\{x_1, x_2, \ldots\}$. This contradicts the assumption that the list contained all numbers in $[0,1]$.

2.3 Ordering of Cardinals and Cantor's Theorem

Definition 2.8. We write $|A| \leq |B|$ (or $A \preceq B$) if there exists an injection $f : A \to B$.

Theorem 2.9 (Cantor-Bernstein)**.** *If $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.*
(i.e., if there exist injections $A \to B$ and $B \to A$, there exists a bijection $A \to B$.)

Additional Proof. Let

$$f : A \to B, \qquad g : B \to A$$

be injections. Define

$$A_0 = A \setminus g(B), \qquad A_{n+1} = g(f(A_n)), \qquad C = \bigcup_{n=0}^{\infty} A_n.$$

Define $h : A \to B$ by

$$h(a) = \begin{cases} f(a), & a \in C, \\ g^{-1}(a), & a \notin C. \end{cases}$$

The second expression is well-defined because if $a \notin C$, then in particular $a \notin A_0$, so $a \in g(B)$; since g is injective, $g^{-1}(a)$ is uniquely defined.

The two parts of the definition have disjoint images. Indeed,

$$f(C) \subset B,$$

while if $a \notin C$ and $g^{-1}(a) = f(c)$ for some $c \in C$, then

$$a = g(f(c)) \in g(f(C)) \subset C,$$

a contradiction. Hence h is injective.

To prove surjectivity, let $b \in B$. If $g(b) \notin C$, then

$$h(g(b)) = g^{-1}(g(b)) = b.$$

If $g(b) \in C$, then $g(b) \notin A_0$, so $g(b) \in A_{n+1} = g(f(A_n))$ for some n. Since g is injective, $b = f(a)$ for some $a \in A_n \subset C$, and therefore

$$h(a) = f(a) = b.$$

Thus h is bijective, so $|A| = |B|$. □

This theorem allows us to order cardinals: $\aleph_0 \leq \mathfrak{c}$. A natural question is whether we can construct cardinalities strictly larger than a given one.

Theorem 2.10 (Cantor's Theorem)**.** *For any set A, $|A| < |2^A|$. That is, there exists an injection $A \to 2^A$, but no surjection from A to 2^A.*

Proof. Clearly $|A| \leq |2^A|$ via the map $x \mapsto \{x\}$.

Assume by contradiction there exists a bijection (or just surjection) $f : A \to 2^A$. Consider the set:

$$E = \{x \in A \mid x \notin f(x)\}.$$

Since $E \subset A$, $E \in 2^A$. Since f is surjective, there exists $a \in A$ such that $f(a) = E$.

- **Case 1:** If $a \in E$, then by definition of E, $a \notin f(a)$. Since $f(a) = E$, this means $a \notin E$. Contradiction.

- **Case 2:** If $a \notin E$, then by definition of E, $a \in f(a)$. Since $f(a) = E$, this means $a \in E$. Contradiction.

Thus, no such mapping exists. □

Remark 2.11. The construction

$$E = \{x \in A \mid x \notin f(x)\}$$

uses the same self-referential "diagonal" idea as Russell's Paradox: membership of x in E is defined by reversing the membership relation $x \in f(x)$.

Corollary 2.12. *By repeatedly taking power sets, we can construct an infinite chain of strictly increasing infinite cardinalities:*

$$\aleph_0 < 2^{\aleph_0} < 2^{2^{\aleph_0}} < \cdots$$

2.4 The Continuum Hypothesis

We know $\aleph_0 < \mathfrak{c}$. Is there a cardinality $\aleph$ such that

$$\aleph_0 < \aleph < \mathfrak{c}?$$

Conjecture 2.13 (Continuum Hypothesis (CH))**.** *There is no set A such that*

$$\aleph_0 < |A| < 2^{\aleph_0}.$$

In other words, $2^{\aleph_0} = \aleph_1$ (the first uncountable cardinal).

Remark 2.14. The Continuum Hypothesis is independent of ZFC (Zermelo-Fraenkel set theory with Choice). It can be neither proved nor disproved within standard set theory (Cohen, 1963).

Example: Cardinality of $C[0,1]$

What is the cardinality of the set of continuous functions on $[0,1]$?

- Since a continuous function is determined by its values on the dense subset $\mathbb{Q} \cap [0,1]$, the map $f \mapsto f|_{\mathbb{Q}}$ is injective.
- Thus,
$$|C[0,1]| \leq |\mathbb{R}^{\mathbb{Q}}| = |(2^{\aleph_0})^{\aleph_0}| = |2^{\aleph_0 \cdot \aleph_0}| = |2^{\aleph_0}| = \mathfrak{c}.$$
- Conversely, considering constant functions, $|\mathbb{R}| \leq |C[0,1]|$.
- By Cantor-Bernstein, the cardinality of $C[0,1]$ is exactly $\mathfrak{c}$.

This is much smaller than the set of all functions $\mathbb{R}^{[0,1]}$, which has cardinality $2^{\mathfrak{c}}$.

Chapter 3

Metric Spaces

3 Metric Spaces

3.1 Definitions and Examples

Definition 3.1 (Metric Space)**.** A metric space is a pair (X, d), where X is a set and $d : X \times X \to \mathbb{R}$ is a function (called a metric or distance) satisfying the following axioms for all $x, y, z \in X$:

1. **Non-negativity:** $d(x, y) \geq 0$.
2. **Identity of Indiscernibles:** $d(x, y) = 0 \iff x = y$.
3. **Symmetry:** $d(x, y) = d(y, x)$.
4. **Triangle Inequality:** $d(x, y) \leq d(x, z) + d(z, y)$.

Example 3.2. 1. **Euclidean Space $\mathbb{R}^n$:**

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

(Euclidean distance or ℓ_2 norm).

2. **Other norms on $\mathbb{R}^n$:**

$$d_1(x, y) = \sum_{i=1}^{n} |x_i - y_i|, \qquad d_\infty(x, y) = \max_{1 \leq i \leq n} |x_i - y_i|.$$

3. **Discrete Metric:** For any set X,

$$d(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

4. **Function Space $C[a, b]$:** The space of continuous functions on $[a, b]$ with the uniform metric:

$$d(f, g) = \max_{x \in [a,b]} |f(x) - g(x)|.$$

Definition 3.3 (Normed Space)**.** A vector space V is a normed space if there is a function

$$\| \cdot \| : V \to \mathbb{R}_{\geq 0}$$

satisfying $\|x\| \geq 0$, $\|\lambda x\| = |\lambda| \|x\|$, and

$$\|x + y\| \leq \|x\| + \|y\|.$$

Every normed space induces a metric

$$d(x, y) = \|x - y\|.$$

3.2 Topology of Metric Spaces

Definition 3.4 (Balls)**.** Let (X, d) be a metric space, $a \in X$, and $R > 0$.

- The **open ball** centered at a with radius R is

$$B_R(a) = \{x \in X : d(x, a) < R\}.$$

- The **closed ball** is
$$\overline{B_R(a)} = \{x \in X : d(x,a) \le R\}.$$

Example 3.5 (Geometry of Balls)**.** In $\mathbb{R}^2$:
- With ℓ_2 norm, balls are disks.
- With ℓ_∞ norm, balls are squares with sides parallel to axes.
- With ℓ_1 norm, balls are "diamonds" (squares rotated by 45°).

Definition 3.6 (Open and Closed Sets)**.**
- A subset $E \subset X$ is **open** if for every $x \in E$, there exists $\varepsilon > 0$ such that $B_\varepsilon(x) \subset E$.
- A subset $E \subset X$ is **closed** if its complement $X \setminus E$ is open.

Theorem 3.7 (Topology Axioms)**.** *1. Arbitrary unions of open sets are open.*
2. Finite intersections of open sets are open.
3. (By duality) Arbitrary intersections of closed sets are closed; finite unions of closed sets are closed.

Additional Proof. For the first statement, let $\{G_\alpha\}_{\alpha \in A}$ be a family of open sets and let
$$x \in \bigcup_{\alpha \in A} G_\alpha.$$
Then $x \in G_{\alpha_0}$ for some α_0. Since G_{α_0} is open, there exists $\varepsilon > 0$ such that
$$B_\varepsilon(x) \subset G_{\alpha_0} \subset \bigcup_{\alpha \in A} G_\alpha.$$
Hence $\bigcup_{\alpha \in A} G_\alpha$ is open.

For the second statement, let $G_1, \dots, G_n$ be open and
$$x \in \bigcap_{i=1}^{n} G_i.$$
For each i, there exists $\varepsilon_i > 0$ such that
$$B_{\varepsilon_i}(x) \subset G_i.$$
Let
$$\varepsilon = \min_{1 \le i \le n} \varepsilon_i > 0.$$
Then
$$B_\varepsilon(x) \subset \bigcap_{i=1}^{n} G_i.$$
Thus the finite intersection is open. The statements for closed sets follow by taking complements and using De Morgan's laws. □

Definition 3.8 (Interior, Exterior, Boundary)**.** Let $E \subset X$.
- x is an **interior point** if $\exists B_r(x) \subset E$.
- x is an **exterior point** if $\exists B_r(x) \subset X \setminus E$.
- x is a **boundary point** if every ball $B_r(x)$ intersects both E and $X \setminus E$.

The set of boundary points is denoted ∂E. Note that ∂E is always closed.

Definition 3.9 (Accumulation Point)**.** A point $x \in X$ is an **accumulation point** (or limit point) of E if every open ball $B_\varepsilon(x)$ contains a point of E distinct from x. Equivalently, there exists a sequence

$$\{x_n\} \subset E \setminus \{x\}$$

such that $x_n \to x$.

Theorem 3.10 (Characterization of Closed Sets)**.** *The following are equivalent:*

1. *E is closed.*
2. *E contains its boundary ($\partial E \subset E$).*
3. *E contains all its accumulation points.*

Proof. (1) $\Rightarrow$ (2): Suppose E is closed. Then $E^c = X \setminus E$ is open. If $x \notin E$, then $x \in E^c$, so there exists $r > 0$ such that

$$B_r(x) \subset E^c.$$

Therefore $B_r(x) \cap E = \varnothing$, so x cannot be a boundary point of E. Hence every boundary point of E lies in E, and therefore

$$\partial E \subset E.$$

(2) $\Rightarrow$ (3): Let x be an accumulation point of E. If $x \notin E$, then $x \in E^c$. Since x is an accumulation point of E, every ball around x contains a point of E. At the same time, every ball around x contains $x \in E^c$. Hence every ball around x intersects both E and E^c, so $x \in \partial E$. Since $\partial E \subset E$, this contradicts $x \notin E$. Thus every accumulation point belongs to E.

(3) $\Rightarrow$ (1): Suppose E contains all its accumulation points. Let $x \in E^c$. Then x is not an accumulation point of E, so there exists $r > 0$ such that

$$B_r(x) \cap (E \setminus \{x\}) = \varnothing.$$

Since $x \notin E$, this gives

$$B_r(x) \cap E = \varnothing,$$

and hence

$$B_r(x) \subset E^c.$$

Therefore E^c is open, so E is closed. □

Definition 3.11 (Closure)**.** The **closure** of E, denoted $\overline{E}$, is the intersection of all closed sets containing E. It satisfies

$$\overline{E} = E \cup \partial E.$$

Remark 3.12. Equivalently,

$$x \in \overline{E} \quad \Longleftrightarrow \quad B_r(x) \cap E \neq \varnothing \text{ for every } r > 0.$$

Thus the closure consists of the points of E together with all points that can be approached arbitrarily closely by points of E.

3.3 Completeness

Definition 3.13 (Cauchy Sequence)**.** A sequence $\{x_n\}$ in a metric space X is called a **Cauchy sequence** (or fundamental sequence) if:

$$\lim_{n,m\to\infty} d(x_n, x_m) = 0.$$

(i.e., $\forall \varepsilon > 0$, $\exists N$ such that $n, m \geq N \Longrightarrow d(x_n, x_m) < \varepsilon$).

Definition 3.14 (Complete Metric Space)**.** A metric space X is **complete** if every Cauchy sequence in X converges to a limit in X.

Example 3.15.
- $\mathbb{R}$ and $\mathbb{R}^n$ are complete.
- $(0, 1)$ is not complete (the sequence $1/n$ is Cauchy but converges to $0 \notin (0, 1)$).
- $\mathbb{Q}$ is not complete.
- $C[a, b]$ with the uniform metric is complete. (The uniform limit of continuous functions is continuous).
- $C[a, b]$ with the integral metric (L^1 norm) is not complete.

Remark 3.16. For the uniform metric on $C[a, b]$,

$$d_\infty(f, g) = \sup_{x\in[a,b]} |f(x) - g(x)|.$$

Thus convergence in this metric is exactly uniform convergence.

Proposition 3.17. *If X is a complete metric space and $E \subset X$ is a closed subset, then (E, d) is complete.*

Proof. Let $\{x_n\} \subset E$ be a Cauchy sequence. Since X is complete, there exists $x \in X$ such that

$$x_n \to x.$$

Because E is closed, it contains all its accumulation points, and hence the limit x belongs to E. Therefore every Cauchy sequence in E converges to a point of E, so (E, d) is complete. □

3.3.1 Banach Fixed Point Theorem

Definition 3.18 (Contraction)**.** Let (X, d) be a metric space. A map $f : X \to X$ is a **contraction** if there exists $q \in [0, 1)$ such that for all $x, y \in X$:

$$d(f(x), f(y)) \leq q \cdot d(x, y).$$

Theorem 3.19 (Banach Fixed Point Theorem)**.** *Let X be a non-empty complete metric space and $f : X \to X$ be a contraction with constant $q < 1$. Then f has a unique fixed point $a \in X$ (i.e., $f(a) = a$). Moreover, for any starting point $x_0 \in X$, the sequence defined by*

$$x_{n+1} = f(x_n)$$

converges to a, with error estimate:

$$d(x_n, a) \leq \frac{q^n}{1 - q} d(x_1, x_0).$$

Sketch. Construct the sequence $x_{n+1} = f(x_n)$. Using the contraction property,

$$d(x_{n+1}, x_n) \leq q^n d(x_1, x_0).$$

By the triangle inequality and geometric series sum, $\{x_n\}$ is Cauchy. Since X is complete, $x_n \to a$. Continuity of f implies $f(a) = a$. Uniqueness follows from

$$d(a, b) = d(f(a), f(b)) \leq qd(a, b) \Longrightarrow d(a, b) = 0.$$

Additional Details. For $m > n$, the triangle inequality gives

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k).$$

Since

$$d(x_{k+1}, x_k) = d(f(x_k), f(x_{k-1})) \leq q\, d(x_k, x_{k-1}),$$

iteration yields

$$d(x_{k+1}, x_k) \leq q^k d(x_1, x_0).$$

Hence

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} q^k d(x_1, x_0) \leq \frac{q^n}{1-q} d(x_1, x_0).$$

Since $q^n \to 0$, the right-hand side tends to 0 as $n, m \to \infty$. Therefore $\{x_n\}$ is Cauchy.

Because X is complete, there exists $a \in X$ such that $x_n \to a$. A contraction is continuous, since

$$d(f(x), f(y)) \leq qd(x, y).$$

Thus

$$f(a) = f\left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} f(x_n) = \lim_{n\to\infty} x_{n+1} = a.$$

Hence a is a fixed point.

If a and b are both fixed points, then

$$d(a, b) = d(f(a), f(b)) \leq qd(a, b).$$

Since $0 \leq q < 1$, this implies

$$(1 - q)d(a, b) \leq 0.$$

Hence $d(a, b) = 0$, so $a = b$.

Finally, letting $m \to \infty$ in

$$d(x_m, x_n) \leq \frac{q^n}{1-q} d(x_1, x_0)$$

gives

$$d(x_n, a) \leq \frac{q^n}{1-q} d(x_1, x_0).$$

□

3.4 Completion of Metric Spaces

Theorem 3.20. *For every metric space (X, d), there exists a complete metric space $(\widetilde{X}, \tilde{d})$ and an isometry*

$$i : X \to \widetilde{X}$$

such that $i(X)$ is dense in $\widetilde{X}$. The space $\widetilde{X}$ is unique up to isometry.

Idea. Define $\widetilde{X}$ as the set of equivalence classes of Cauchy sequences in X, where

$$\{x_n\} \sim \{y_n\} \iff d(x_n, y_n) \to 0.$$

The metric is defined by

$$\tilde{d}([\{x_n\}], [\{y_n\}]) = \lim_{n\to\infty} d(x_n, y_n).$$

Remark 3.21. The original space X is embedded into $\widetilde{X}$ by identifying every point $x \in X$ with the constant Cauchy sequence

$$x, x, x, \ldots.$$

Thus

$$i(x) = [\{x, x, x, \ldots\}].$$

The statement that $i(X)$ is dense in $\widetilde{X}$ means that every element of the completion can be approximated arbitrarily well by points coming from the original space X.

Additional Details of the Construction. We briefly verify why the construction above works.

First, define an equivalence relation on the set of all Cauchy sequences in X by

$$\{x_n\} \sim \{y_n\} \iff d(x_n, y_n) \to 0.$$

Let $\widetilde{X}$ be the set of equivalence classes.

For two classes $[\{x_n\}]$ and $[\{y_n\}]$, define

$$\tilde{d}([\{x_n\}], [\{y_n\}]) = \lim_{n\to\infty} d(x_n, y_n).$$

The limit exists because

$$|d(x_n, y_n) - d(x_m, y_m)| \le d(x_n, x_m) + d(y_n, y_m),$$

and both $\{x_n\}$ and $\{y_n\}$ are Cauchy. Hence the real sequence $\{d(x_n, y_n)\}$ is Cauchy and therefore convergent.

The value of $\tilde{d}$ does not depend on the representatives of the equivalence classes. Indeed, if

$$x_n \sim x'_n, \qquad y_n \sim y'_n,$$

then

$$|d(x_n, y_n) - d(x'_n, y'_n)| \le d(x_n, x'_n) + d(y_n, y'_n) \to 0.$$

Define

$$i : X \to \widetilde{X}, \qquad i(x) = [\{x, x, x, \ldots\}].$$

Then

$$\tilde{d}(i(x), i(y)) = \lim_{n\to\infty} d(x, y) = d(x, y),$$

so i is an isometry.

If $[\{x_n\}] \in \widetilde{X}$, then

$$i(x_n) \to [\{x_n\}]$$

in $\widetilde{X}$. Therefore $i(X)$ is dense in $\widetilde{X}$. One can further show that every Cauchy sequence of equivalence classes converges in $\widetilde{X}$, so $\widetilde{X}$ is complete. □

Chapter 4

Compactness

4 Compactness

4.1 Definitions of Compactness

Definition 4.1. Let K be a metric space. An **open covering** of K is a collection $\{G_\alpha\}_{\alpha\in A}$ of open sets such that

$$K \subset \bigcup_{\alpha\in A} G_\alpha.$$

Definition 4.2 (Compactness)**.** A metric space K is called **compact** if every open covering admits a finite subcovering. i.e., $\exists\alpha_1,\ldots,\alpha_n$ such that

$$K \subset \bigcup_{i=1}^{n} G_{\alpha_i}.$$

Theorem 4.3 (Heine-Borel)**.** *A subset $K \subset \mathbb{R}^n$ is compact if and only if it is closed and bounded.*

Additional Proof. We prove the two directions separately.

Compact ⇒ bounded. Fix $x_0 \in \mathbb{R}^n$. The collection

$$\{B_m(x_0)\}_{m=1}^{\infty}$$

is an open covering of K. Since K is compact, there exist

$$m_1,\ldots,m_N$$

such that

$$K \subset \bigcup_{i=1}^{N} B_{m_i}(x_0).$$

Let

$$M = \max_{1\le i\le N} m_i.$$

Then

$$K \subset B_M(x_0),$$

so K is bounded.

Compact ⇒ closed. Let $x \notin K$. For every $y \in K$, define

$$r_y = \frac{d(x,y)}{2} > 0.$$

Then

$$\{B_{r_y}(y)\}_{y\in K}$$

is an open covering of K. By compactness, there exist

$$y_1,\ldots,y_N \in K$$

such that

$$K \subset \bigcup_{i=1}^{N} B_{r_{y_i}}(y_i).$$

Let

$$r = \min_{1\le i\le N} r_{y_i} > 0.$$

We claim that

$$B_r(x) \cap K = \varnothing.$$

Indeed, if $z \in B_r(x) \cap K$, then $z \in B_{r_{y_i}}(y_i)$ for some i. Hence

$$d(x, y_i) \leq d(x, z) + d(z, y_i) < r + r_{y_i} \leq 2r_{y_i} = d(x, y_i),$$

a contradiction. Therefore

$$B_r(x) \subset K^c.$$

Thus K^c is open and K is closed.

Closed and bounded $\Rightarrow$ compact. This is the classical Heine–Borel theorem in $\mathbb{R}^n$. One standard proof uses the fact that bounded subsets of $\mathbb{R}^n$ are totally bounded and closed subsets of the complete space $\mathbb{R}^n$ are complete. By Theorem 4.9 below, complete and totally bounded metric spaces are compact. □

Example 4.4.
- $[a, b]$ is compact.
- (a, b) is not compact (Cover by $(a + 1/n, b - 1/n)$, no finite subcover).
- $\mathbb{R}$ is not compact (unbounded).

Proposition 4.5. *1. A closed subset of a compact space is compact.*
2. A compact subset of any metric space is closed.

Additional Proof. For (1), let $F \subset K$ be closed and suppose

$$F \subset \bigcup_{\alpha \in A} G_\alpha$$

is an open covering of F. Since F is closed in K, the set $K \setminus F$ is open. Therefore

$$K \subset (K \setminus F) \cup \bigcup_{\alpha \in A} G_\alpha$$

is an open covering of K. Since K is compact, finitely many of these sets cover K. Removing $K \setminus F$ if necessary leaves finitely many G_α covering F. Hence F is compact.

For (2), let K be compact and let $x \notin K$. The same argument used in the proof of Heine–Borel above gives an open ball $B_r(x)$ disjoint from K. Thus K^c is open and K is closed. □

Theorem 4.6 (Finite Intersection Property)**.** *A space K is compact if and only if for every family of closed subsets $\{F_\alpha\}$ satisfying the finite intersection property (any finite sub-collection has non-empty intersection), the total intersection*

$$\bigcap_{\alpha} F_\alpha$$

is non-empty.

Proof. Suppose first that K is compact. Assume

$$\bigcap_{\alpha \in A} F_\alpha = \varnothing.$$

Taking complements,

$$K = \left(\bigcap_{\alpha \in A} F_\alpha \right)^c = \bigcup_{\alpha \in A} F_\alpha^c.$$

Since each F_α is closed, each F_α^c is open. Thus

$$\{F_\alpha^c\}_{\alpha \in A}$$

is an open covering of K. By compactness, there exist

$$\alpha_1, \ldots, \alpha_n$$

such that

$$K = \bigcup_{i=1}^{n} F_{\alpha_i}^c.$$

Taking complements,

$$\bigcap_{i=1}^{n} F_{\alpha_i} = \varnothing,$$

contradicting the finite intersection property.

Conversely, suppose the finite intersection property holds for every family of closed subsets. Let

$$K \subset \bigcup_{\alpha \in A} G_\alpha$$

be an open covering. Suppose no finite subcollection covers K. Define

$$F_\alpha = K \setminus G_\alpha.$$

Each F_α is closed, and because no finite subcollection of $\{G_\alpha\}$ covers K, every finite intersection

$$F_{\alpha_1} \cap \cdots \cap F_{\alpha_n}$$

is non-empty. Thus $\{F_\alpha\}$ satisfies the finite intersection property, so

$$\bigcap_{\alpha \in A} F_\alpha \neq \varnothing.$$

But

$$\bigcap_{\alpha \in A} F_\alpha = K \setminus \bigcup_{\alpha \in A} G_\alpha = \varnothing,$$

a contradiction. Therefore the open covering has a finite subcover. □

4.2 Sequential Compactness and Total Boundedness

Definition 4.7 (Sequential Compactness)**.** A metric space K is **sequentially compact** if every sequence $\{x_n\} \subset K$ has a convergent subsequence with limit in K.

Definition 4.8 (ε-net and Total Boundedness)**.** Let $\varepsilon > 0$. A finite set $\{a_1, \ldots, a_n\} \subset K$ is an ε-net if

$$K \subset \bigcup_{i=1}^{n} B_\varepsilon(a_i).$$

K is **totally bounded** if for every $\varepsilon > 0$, it admits a finite ε-net.

Theorem 4.9 (Fundamental Theorem of Metric Compactness)**.** *Let K be a metric space. The following are equivalent:*

1. *K is compact (Heine-Borel).*
2. *K is sequentially compact (Bolzano-Weierstrass).*

3. K is complete and totally bounded.

Proof. We prove

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1).$$

(1) $\Rightarrow$ (2)**.** Suppose K is compact and let $\{x_n\} \subset K$. Assume that $\{x_n\}$ has no convergent subsequence.

Then no point $y \in K$ can be an accumulation point of the sequence. Hence for every $y \in K$, there exists an open neighborhood U_y containing only finitely many terms of the sequence $\{x_n\}$.

The family

$$\{U_y\}_{y \in K}$$

is an open covering of K. By compactness, there exist

$$y_1, \ldots, y_N$$

such that

$$K \subset \bigcup_{i=1}^{N} U_{y_i}.$$

Each U_{y_i} contains only finitely many terms x_n, so their finite union contains only finitely many terms of the sequence. But every $x_n \in K$, a contradiction. Thus $\{x_n\}$ has a convergent subsequence with limit in K.

(2) $\Rightarrow$ (3)**: completeness.** Let $\{x_n\}$ be a Cauchy sequence in K. By sequential compactness, it has a convergent subsequence

$$x_{n_k} \to x \in K.$$

Since $\{x_n\}$ is Cauchy, the whole sequence converges to the same limit x. Indeed, given $\varepsilon > 0$, choose N such that

$$m, n \geq N \Longrightarrow d(x_m, x_n) < \frac{\varepsilon}{2}.$$

Choose k sufficiently large that

$$n_k \geq N \qquad \text{and} \qquad d(x_{n_k}, x) < \frac{\varepsilon}{2}.$$

Then for $n \geq N$,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon.$$

Hence K is complete.

(2) $\Rightarrow$ (3)**: total boundedness.** Suppose K is not totally bounded. Then there exists $\varepsilon > 0$ such that no finite collection of ε-balls covers K.

Choose $x_1 \in K$. Since $B_\varepsilon(x_1)$ does not cover K, choose

$$x_2 \in K \setminus B_\varepsilon(x_1).$$

After choosing $x_1, \ldots, x_n$, the union

$$\bigcup_{i=1}^{n} B_\varepsilon(x_i)$$

does not cover K, so choose

$$x_{n+1} \in K \setminus \bigcup_{i=1}^{n} B_\varepsilon(x_i).$$

Thus

$$d(x_n, x_m) \geq \varepsilon \qquad (n \neq m).$$

Therefore $\{x_n\}$ has no Cauchy subsequence and hence no convergent subsequence, contradicting sequential compactness. Thus K is totally bounded.

(3) $\Rightarrow$ (1)**.** Assume K is complete and totally bounded, and let $\mathcal{U}$ be an open covering of K. Suppose no finite subcollection of $\mathcal{U}$ covers K.

Since K is totally bounded, there is a finite 1/2-net

$$K \subset \bigcup_{i=1}^{N_1} B_{1/2}(x_i).$$

At least one of these balls, say B_1, has the property that

$$K \cap \overline{B_1}$$

cannot be covered by finitely many elements of $\mathcal{U}$.

Next choose a finite 1/4-net for $K \cap \overline{B_1}$. Again one of the corresponding closed balls, say B_2, cannot be finitely covered by $\mathcal{U}$. Continue inductively, obtaining nested non-empty closed sets

$$B_1 \supset B_2 \supset \cdots$$

with

$$\operatorname{diam}(B_n) \to 0.$$

Choose $x_n \in B_n$. Since the sets are nested and their diameters tend to 0, $\{x_n\}$ is Cauchy. Completeness implies

$$x_n \to x^* \in K.$$

Since $\mathcal{U}$ covers K, there exists $U \in \mathcal{U}$ such that

$$x^* \in U.$$

Because U is open, there exists $r > 0$ such that

$$B_r(x^*) \subset U.$$

For sufficiently large n, the set B_n is contained in $B_r(x^*)$, hence

$$B_n \subset U.$$

But then B_n is covered by the single set U, contradicting the construction. Therefore $\mathcal{U}$ has a finite subcover, and K is compact. □

Remark 4.10. In infinite-dimensional spaces (like function spaces), "bounded and closed" is not sufficient for compactness. "Total boundedness" is the correct strengthening of boundedness required.

Remark 4.11. Every totally bounded set is bounded, but the converse need not hold. In finite-dimensional Euclidean spaces, boundedness implies total boundedness, which is one reason why the Heine–Borel theorem takes the simpler form "closed and bounded" in $\mathbb{R}^n$.

4.3 Functions on Compact Spaces

Theorem 4.12. *Let $f : X \to Y$ be a continuous map between metric spaces. If $K \subset X$ is compact, then the image $f(K)$ is compact in Y.*

Proof. Let $\{V_\alpha\}_{\alpha \in A}$ be an open covering of $f(K)$. Since f is continuous,

$$f^{-1}(V_\alpha)$$

is open in X for every α, and

$$K \subset \bigcup_{\alpha \in A} f^{-1}(V_\alpha).$$

Since K is compact, there exist $\alpha_1, \ldots, \alpha_n$ such that

$$K \subset \bigcup_{i=1}^{n} f^{-1}(V_{\alpha_i}).$$

Applying f,

$$f(K) \subset \bigcup_{i=1}^{n} V_{\alpha_i}.$$

Thus $f(K)$ is compact. □

Corollary 4.13 (Extreme Value Theorem)**.** *Let $f : K \to \mathbb{R}$ be a continuous function on a compact set K. Then f is bounded and attains its maximum and minimum values.*

Additional Proof. By the previous theorem, $f(K)$ is compact in $\mathbb{R}$. By the Heine–Borel theorem, $f(K)$ is closed and bounded.

Since $f(K)$ is bounded, f is bounded. Let

$$M = \sup f(K).$$

There exists a sequence $\{y_n\} \subset f(K)$ such that

$$y_n \to M.$$

Since $f(K)$ is closed, $M \in f(K)$. Hence there exists $x_{\max} \in K$ such that

$$f(x_{\max}) = M.$$

Similarly, f attains its minimum. □

Theorem 4.14 (Uniform Continuity)**.** *If $f : K \to Y$ is continuous and K is compact, then f is uniformly continuous.*

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y \in K : d_X(x, y) < \delta \Longrightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Proof. Suppose f is not uniformly continuous. Then there exists $\varepsilon_0 > 0$ such that for every $n \in \mathbb{N}$ there exist $x_n, y_n \in K$ satisfying

$$d_X(x_n, y_n) < \frac{1}{n}, \qquad d_Y(f(x_n), f(y_n)) \geq \varepsilon_0.$$

Since K is compact, it is sequentially compact. Thus $\{x_n\}$ has a convergent subsequence

$$x_{n_k} \to x \in K.$$

Because

$$d_X(x_{n_k}, y_{n_k}) < \frac{1}{n_k} \to 0,$$

we also have

$$y_{n_k} \to x.$$

By continuity of f,

$$f(x_{n_k}) \to f(x), \qquad f(y_{n_k}) \to f(x).$$

Therefore

$$d_Y(f(x_{n_k}), f(y_{n_k})) \to 0,$$

contradicting

$$d_Y(f(x_{n_k}), f(y_{n_k})) \geq \varepsilon_0.$$

Hence f is uniformly continuous. □

4.4 The Arzelà-Ascoli Theorem

This theorem characterizes compact sets in the space of continuous functions $C(K)$.

Definition 4.15 (Equicontinuity)**.** A family of functions $\mathcal{F} \subset C(K)$ is **equicontinuous** if:

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y \in K,\ \forall f \in \mathcal{F} : d(x, y) < \delta \Longrightarrow |f(x) - f(y)| < \varepsilon.$$

Remark 4.16. The crucial point is that the same δ works simultaneously for every $f \in \mathcal{F}$.

Definition 4.17 (Uniform Boundedness)**.** A family $\mathcal{F}$ is **uniformly bounded** if $\exists M > 0$ such that

$$|f(x)| \leq M$$

for all $x \in K$ and $f \in \mathcal{F}$.

Theorem 4.18 (Arzelà-Ascoli)**.** *Let K be a compact metric space. A subset $\mathcal{F} \subset C(K)$ is precompact (i.e., every sequence in $\mathcal{F}$ has a uniformly convergent subsequence) if and only if $\mathcal{F}$ is:*

1. *Uniformly bounded.*
2. *Equicontinuous.*

Proof Sketch (Diagonalization). **Necessity:** Follows from general properties of compact spaces (total boundedness implies equicontinuity for function spaces).

Sufficiency:

1. Let $\{f_n\} \subset \mathcal{F}$. Since K is compact, it is separable. Let

   $$D = \{x_1, x_2, \ldots\}$$

   be a countable dense subset of K.
2. Since $\{f_n(x_1)\}$ is bounded, extract a subsequence $f_{1,k}$ converging at x_1. From this, extract $f_{2,k}$ converging at x_2, and so on.
3. Consider the diagonal sequence

   $$g_k = f_{k,k}.$$

   It converges pointwise on D.
4. Use the $\varepsilon/3$-argument and equicontinuity to show that g_k is actually a Cauchy sequence in the uniform norm ($C(K)$). Since $C(K)$ is complete, g_k converges uniformly.

Additional Details for Sufficiency. Let $\{g_k\}$ be the diagonal subsequence constructed above. We show that it is uniformly Cauchy.

Fix $\varepsilon > 0$. By equicontinuity, there exists $\delta > 0$ such that for every $f \in \mathcal{F}$,

$$d(x, y) < \delta \Longrightarrow |f(x) - f(y)| < \frac{\varepsilon}{3}.$$

Since K is compact, it is totally bounded. Hence there exist finitely many points

$$x_{i_1}, \ldots, x_{i_N} \in D$$

such that

$$K \subset \bigcup_{j=1}^{N} B_\delta(x_{i_j}).$$

Because $g_k(x_{i_j})$ converges for every j, there exists N_0 such that for all $m, n \geq N_0$ and all $j = 1, \ldots, N$,

$$|g_m(x_{i_j}) - g_n(x_{i_j})| < \frac{\varepsilon}{3}.$$

Now let $x \in K$. Choose j such that

$$d(x, x_{i_j}) < \delta.$$

Then

$$\begin{aligned} |g_m(x) - g_n(x)| &\leq |g_m(x) - g_m(x_{i_j})| + |g_m(x_{i_j}) - g_n(x_{i_j})| \\ &\quad + |g_n(x_{i_j}) - g_n(x)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Therefore

$$\|g_m - g_n\|_\infty < \varepsilon$$

for all $m, n \geq N_0$. Thus $\{g_k\}$ is Cauchy in the uniform norm.

Since $C(K)$ is complete, there exists $g \in C(K)$ such that

$$\|g_k - g\|_\infty \to 0.$$

Hence g_k converges uniformly. □

Chapter 5

Approximation and Density

5 Approximation and Density

5.1 Density and Separability

Definition 5.1 (Dense Set)**.** Let (X, d) be a metric space. A subset $E \subset X$ is called dense in X if

$$\overline{E} = X.$$

Equivalently, for every $x \in X$ and every $\varepsilon > 0$, the ball $B_\varepsilon(x)$ contains an element of E (i.e.,

$$B_\varepsilon(x) \cap E \neq \varnothing).$$

Definition 5.2 (Separable Space)**.** A metric space X is called separable if it contains a countable dense subset.

Example 5.3.
1. $\mathbb{Q}$ is dense in $\mathbb{R}$. Thus $\mathbb{R}$ is separable.
2. $\mathbb{Q}^n$ is dense in $\mathbb{R}^n$. Thus $\mathbb{R}^n$ is separable.
3. **Polynomials:** The Weierstrass Approximation Theorem states that the set of polynomials $\mathcal{P}$ is dense in $C[a, b]$ equipped with the uniform norm. Since polynomials with rational coefficients form a countable set, $C[a, b]$ is separable.
4. L^p **spaces:** $L^p(\mathbb{R}^n)$ is separable for $1 \leq p < \infty$.
5. **Counterexample:** $L^\infty[0, 1]$ is not separable. The uncountable collection of functions

   $$\{\chi_{[0,t]} \mid t \in (0, 1)\}$$

   satisfies

   $$\|\chi_{[0,t]} - \chi_{[0,s]}\|_\infty = 1$$

   for $t \neq s$, so no countable dense set can exist.

Remark 5.4. For $C[a, b]$, the uniform norm is

$$\|f - g\|_\infty = \sup_{x \in [a,b]} |f(x) - g(x)|.$$

Thus density of the polynomials in $C[a, b]$ means that every continuous function can be approximated arbitrarily well, uniformly on $[a, b]$, by polynomials.

Proposition 5.5. *Every compact metric space is separable.*

Proof. For every $n \in \mathbb{N}$, let S_n be a finite $1/n$-net. The union

$$S = \bigcup_{n=1}^{\infty} S_n$$

is a countable dense subset. □

Remark 5.6. The existence of the finite $1/n$-net follows from compactness, since compact metric spaces are totally bounded.

5.2 The Baire Category Theorem

This theorem relates the topological size of a set (density) to the completeness of the space.

Definition 5.7 (Nowhere Dense)**.** A subset $E \subset X$ is called **nowhere dense** if its closure has an empty interior, i.e.,

$$(\overline{E})^\circ = \varnothing.$$

Geometrically, this means E is "full of holes": every open set contains a ball disjoint from E.

Example 5.8.
- $\mathbb{Z}$ is nowhere dense in $\mathbb{R}$.
- The Cantor set C is nowhere dense in $[0, 1]$ (it is closed and contains no intervals).
- $\mathbb{Q}$ is not nowhere dense in $\mathbb{R}$ (its closure is $\mathbb{R}$, which has non-empty interior).

Proposition 5.9. *The complement of an open dense set is nowhere dense.*

Proof. Let $G \subset X$ be open and dense, and define

$$F = X \setminus G.$$

Since G is open, F is closed, so

$$\overline{F} = F.$$

We claim that

$$F^\circ = \varnothing.$$

Indeed, if $F^\circ \neq \varnothing$, then there exists a non-empty open set U such that

$$U \subset F = X \setminus G.$$

Hence

$$U \cap G = \varnothing,$$

contradicting the fact that G is dense, since every non-empty open set must intersect G. Therefore

$$(\overline{F})^\circ = F^\circ = \varnothing,$$

so F is nowhere dense. □

Definition 5.10 (Categories)**.**
- A set is of **First Category** (or meager) if it is a countable union of nowhere dense sets.
- A set is of **Second Category** if it is not of the first category.

Remark 5.11. A set of first category need not be "small" in the sense of density. For example,

$$\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\}$$

is a countable union of nowhere dense singleton sets, so $\mathbb{Q}$ is of first category in $\mathbb{R}$, but $\mathbb{Q}$ is nevertheless dense in $\mathbb{R}$.

Theorem 5.12 (Baire Category Theorem)**.** *Let X be a complete metric space. Then X is not the union of a countable family of nowhere dense sets. Consequently, if $X \neq \varnothing$, X is of Second Category in itself.*

Corollary 5.13. *In a complete metric space, the intersection of countably many open dense sets is dense.*

Proof of Baire Category Theorem. Let $\{E_n\}_{n=1}^{\infty}$ be a sequence of nowhere dense sets. We want to show

$$U = X \setminus \bigcup E_n \neq \varnothing.$$

Since E_1 is nowhere dense, $\overline{E_1}$ has empty interior. Thus, given any open ball B_0, there exists a closed ball

$$\overline{B_1} \subset B_0$$

such that

$$\overline{B_1} \cap E_1 = \varnothing$$

(choose radius $r_1 < 1$).

Inductively, suppose we have chosen a closed ball $\overline{B_n}$. Since E_{n+1} is nowhere dense, there exists a closed ball

$$\overline{B_{n+1}} \subset B_n$$

such that

$$\overline{B_{n+1}} \cap E_{n+1} = \varnothing,$$

and we can ensure radius

$$r_{n+1} < \frac{1}{n+1}.$$

We obtain a nested sequence of closed balls

$$\overline{B_1} \supset \overline{B_2} \supset \cdots$$

with radii tending to 0. By the completeness of X (Cantor's Intersection Theorem), there exists a unique point

$$x \in \bigcap_{n=1}^{\infty} \overline{B_n}.$$

By construction,

$$x \notin E_n$$

for any n. Therefore,

$$x \notin \bigcup E_n,$$

so the union cannot be all of X. □

Remark 5.14. The key use of nowhere density is that inside every non-empty open ball one can find a smaller ball avoiding $\overline{E_n}$. Thus the construction repeatedly "escapes" from

$$E_1, E_2, E_3, \ldots$$

while keeping the balls nested.

Proof of Corollary 5.9. Let

$$G = \bigcap_{n=1}^{\infty} G_n,$$

where every G_n is open and dense. We show that G is dense.

For each n, let

$$F_n = X \setminus G_n.$$

Since G_n is open, F_n is closed. Since G_n is dense,

$$F_n^\circ = \varnothing,$$

so each F_n is nowhere dense.

Now

$$X \setminus G = X \setminus \bigcap_{n=1}^{\infty} G_n = \bigcup_{n=1}^{\infty} F_n.$$

By the Baire Category Theorem, this countable union of nowhere dense sets cannot contain any non-empty open subset of X.

Therefore every non-empty open set intersects G, and hence G is dense. □

5.3 The Stone-Weierstrass Theorem

This is the vast generalization of approximating continuous functions by polynomials.

Definition 5.15 (Algebra of Functions)**.** A subset $\mathcal{A} \subset C(K)$ (where K is a compact metric space) is called an **algebra** if it is a linear subspace closed under multiplication:

$$f, g \in \mathcal{A} \Longrightarrow f \cdot g \in \mathcal{A}.$$

Remark 5.16. Thus an algebra of functions is closed under addition, scalar multiplication, and multiplication of functions.

Theorem 5.17 (Stone-Weierstrass)**.** *Let K be a compact metric space. Let $\mathcal{A}$ be a subalgebra of $C(K, \mathbb{R})$ with the following properties:*

1. *$\mathcal{A}$ **separates points**: $\forall x \neq y \in K$, $\exists f \in \mathcal{A}$ such that*

$$f(x) \neq f(y).$$

2. *$\mathcal{A}$ **vanishes nowhere**: $\forall x \in K$, $\exists f \in \mathcal{A}$ such that*

$$f(x) \neq 0$$

(or simply, the constant function $1 \in \mathcal{A}$).

Then $\mathcal{A}$ is dense in $C(K)$ (in the uniform norm). That is,

$$\overline{\mathcal{A}} = C(K).$$

Remark 5.18. "Separates points" means that any two distinct points of K can be distinguished by at least one function in $\mathcal{A}$. "Vanishes nowhere" means that there is no point of K at which every function in $\mathcal{A}$ is zero.

Example 5.19.

- **Polynomials on $[a, b]$:**

$$\mathcal{A} = \operatorname{span}\{1, x, x^2, \ldots\}.$$

Separates points ($f(x) = x$) and contains 1.

$$\Longrightarrow \text{Weierstrass Approximation Theorem.}$$

- **Trigonometric Polynomials on circle:** Must use complex version or

$$\{1, \sin nx, \cos nx\}.$$

- **Functions on $\mathbb{R}$:** The theorem fails if K is not compact (e.g., polynomials are not dense in $C_b(\mathbb{R})$).

Proof Outline. Let $\overline{\mathcal{A}}$ be the closure of $\mathcal{A}$. It is a closed algebra.
Step 1: Lattice Property. We claim that if $f \in \overline{\mathcal{A}}$, then

$$|f| \in \overline{\mathcal{A}}.$$

This relies on approximating the function

$$t \mapsto |t|$$

by polynomials $P_n(t)$ on $[-M, M]$ (using Taylor series of $\sqrt{t^2 + \varepsilon}$).

Since

$$\max(f, g) = \frac{f + g + |f - g|}{2}$$

and

$$\min(f, g) = \frac{f + g - |f - g|}{2},$$

$\overline{\mathcal{A}}$ is closed under max and min operations (it is a vector lattice).
Step 2: Pointwise Interpolation. For any distinct $x, y \in K$ and values $\alpha, \beta \in \mathbb{R}$, there exists

$$f \in \overline{\mathcal{A}}$$

such that

$$f(x) = \alpha, \qquad f(y) = \beta.$$

(This follows from the separation property and solving a simple linear system).
Step 3: Global Approximation. Let $h \in C(K)$ and $\varepsilon > 0$. Fix $x \in K$. For every $y \in K$, use Step 2 to find

$$f_y \in \overline{\mathcal{A}}$$

such that

$$f_y(x) = h(x) \qquad \text{and} \qquad f_y(y) = h(y).$$

By continuity,

$$f_y > h - \varepsilon$$

in a neighborhood of y. Since K is compact, cover K with finite neighborhoods and take

$$g_x = \max(f_{y_1}, \ldots, f_{y_k}).$$

Then

$$g_x > h - \varepsilon$$

on all of K, and

$$g_x(x) = h(x).$$

Now,

$$g_x < h + \varepsilon$$

in a neighborhood of x. Cover K again and take

$$F = \min(g_{x_1}, \ldots, g_{x_m}).$$

Then

$$h - \varepsilon < F < h + \varepsilon.$$

Since

$$F \in \overline{\mathcal{A}}$$

(lattice property), h is approximated by elements of $\mathcal{A}$.

Additional Details for Step 2. Let $x \neq y$. Since $\mathcal{A}$ separates points, there exists $g \in \mathcal{A}$ such that

$$g(x) \neq g(y).$$

Assuming the constant function $1 \in \mathcal{A}$, consider

$$f = a + bg.$$

We want

$$a + bg(x) = \alpha, \qquad a + bg(y) = \beta.$$

Since $g(x) \neq g(y)$, the system has a unique solution:

$$b = \frac{\alpha - \beta}{g(x) - g(y)}, \qquad a = \alpha - bg(x).$$

Since $\mathcal{A}$ is a linear space and $1, g \in \mathcal{A}$, we have

$$f \in \mathcal{A} \subset \overline{\mathcal{A}}.$$

Thus the required interpolation function exists. □

Additional Details for Step 3. Fix $h \in C(K)$ and $\varepsilon > 0$. For a fixed $x \in K$, Step 2 gives, for every $y \in K$, a function $f_y \in \overline{\mathcal{A}}$ such that

$$f_y(x) = h(x), \qquad f_y(y) = h(y).$$

Since

$$f_y(y) = h(y) > h(y) - \varepsilon,$$

continuity of $f_y - h$ implies that there exists a neighborhood U_y of y such that

$$f_y(z) > h(z) - \varepsilon$$

for all $z \in U_y$.

The family $\{U_y\}_{y \in K}$ covers K. By compactness, choose

$$y_1, \ldots, y_k$$

such that

$$K \subset U_{y_1} \cup \cdots \cup U_{y_k}.$$

Define

$$g_x = \max(f_{y_1}, \ldots, f_{y_k}).$$

By the lattice property,

$$g_x \in \overline{\mathcal{A}}.$$

For every $z \in K$, at least one U_{y_i} contains z, hence

$$g_x(z) \geq f_{y_i}(z) > h(z) - \varepsilon.$$

Moreover,

$$g_x(x) = h(x),$$

because every $f_{y_i}(x) = h(x)$.

Therefore, by continuity, there exists a neighborhood V_x of x such that

$$g_x(z) < h(z) + \varepsilon$$

for $z \in V_x$.

The family $\{V_x\}_{x \in K}$ covers K. By compactness, choose

$$x_1, \ldots, x_m$$

such that

$$K \subset V_{x_1} \cup \cdots \cup V_{x_m}.$$

Set

$$F = \min(g_{x_1}, \ldots, g_{x_m}).$$

Again, the lattice property implies

$$F \in \overline{\mathcal{A}}.$$

Since every $g_{x_i} > h - \varepsilon$ on K,

$$F > h - \varepsilon.$$

For each $z \in K$, choose i such that $z \in V_{x_i}$. Then

$$F(z) \leq g_{x_i}(z) < h(z) + \varepsilon.$$

Hence

$$|F(z) - h(z)| < \varepsilon$$

for all $z \in K$, so

$$\|F - h\|_\infty < \varepsilon.$$

Therefore

$$h \in \overline{\mathcal{A}}.$$

Since $h \in C(K)$ was arbitrary,

$$\overline{\mathcal{A}} = C(K).$$

□

Chapter 6

Construction of Measures

6 Construction of Measures

The goal of this chapter is to rigorously define a "volume" or "measure" for subsets of $\mathbb{R}^n$. We start with simple sets (intervals/cells) and extend the notion of volume to a larger class of sets (σ-algebra) using the Carathéodory extension process.

6.1 Rings, Algebras, and Semi-rings

Definition 6.1 (Ring of Sets)**.** A non-empty collection of sets $\mathcal{R}$ is called a Ring if it is closed under:

1. **Union:** $A, B \in \mathcal{R} \implies A \cup B \in \mathcal{R}$.
2. **Difference:** $A, B \in \mathcal{R} \implies A \setminus B \in \mathcal{R}$.

Note: Since

$$A \cap B = A \setminus (A \setminus B),$$

a ring is also closed under finite intersections. Since

$$\varnothing = A \setminus A,$$

$\varnothing \in \mathcal{R}$.

Remark 6.2. Thus a ring is closed under finite unions, finite intersections, and differences. It does not necessarily contain the universal set.

Definition 6.3 (Algebra)**.** A ring $\mathcal{R}$ is called an Algebra if the universal set $\Omega \in \mathcal{R}$. Consequently, an algebra is closed under complementation:

$$A^c = \Omega \setminus A \in \mathcal{R}.$$

Definition 6.4 (σ-Algebra)**.** An algebra $\mathcal{A}$ is a σ-algebra if it is closed under countable unions:

$$A_j \in \mathcal{A},\ \forall j \in \mathbb{N} \implies \bigcup_{j=1}^{\infty} A_j \in \mathcal{A}.$$

Remark 6.5. A σ-algebra is therefore closed under countable intersections as well, since by De Morgan's law,

$$\bigcap_{j=1}^{\infty} A_j = \left(\bigcup_{j=1}^{\infty} A_j^c \right)^c.$$

This is the natural collection of sets on which a measure is defined.

Often, the collection of sets we start with (like rectangles) is not a ring. We need a weaker structure.

Definition 6.6 (Semi-ring)**.** A collection $\mathcal{S}$ is called a Semi-ring if:

1. $\varnothing \in \mathcal{S}$.
2. Closed under intersection:

$$A, B \in \mathcal{S} \implies A \cap B \in \mathcal{S}.$$

3. Difference property: If $A, B \in \mathcal{S}$, then $A \setminus B$ can be represented as a finite disjoint union of elements in $\mathcal{S}$:

$$A \setminus B = \bigsqcup_{k=1}^{n} S_k, \qquad S_k \in \mathcal{S}.$$

Remark 6.7. The essential difference from a ring is that $A \setminus B$ need not itself belong to $\mathcal{S}$; it only needs to be expressible as a finite disjoint union of sets belonging to $\mathcal{S}$.

Example 6.8 (Cells in $\mathbb{R}^n$)**.** Let $\mathcal{S}$ be the collection of semi-open intervals $[a, b)$ in $\mathbb{R}$.

- Intersection: $[a, b) \cap [c, d)$ is either empty or an interval

$$[\max(a, c), \min(b, d)).$$

- Difference: $[a, b) \setminus [c, d)$ results in at most two semi-open intervals.

Thus, $\mathcal{S}$ is a semi-ring. Similarly, the collection of "cells" (products of intervals) in $\mathbb{R}^n$ is a semi-ring.

Proposition 6.9. *Let $\mathcal{S}$ be a semi-ring. The collection of all finite disjoint unions of sets from $\mathcal{S}$ forms a Ring, denoted $\mathcal{R}(\mathcal{S})$, generated by $\mathcal{S}$.*

Additional Proof. Define

$$\mathcal{R}(\mathcal{S}) = \left\{ \bigsqcup_{i=1}^{n} A_i : A_i \in \mathcal{S},\ n < \infty \right\}.$$

Clearly $\varnothing \in \mathcal{R}(\mathcal{S})$.

Take

$$A = \bigsqcup_{i=1}^{m} A_i, \qquad B = \bigsqcup_{j=1}^{n} B_j.$$

Using the intersection and difference properties of the semi-ring, the sets can be decomposed into finitely many mutually disjoint pieces from $\mathcal{S}$. Consequently, both

$$A \cup B \qquad \text{and} \qquad A \setminus B$$

can be written as finite disjoint unions of elements of $\mathcal{S}$. Hence

$$A \cup B,\ A \setminus B \in \mathcal{R}(\mathcal{S}),$$

so $\mathcal{R}(\mathcal{S})$ is a ring. □

6.2 Measures on Semi-rings and Extension

Definition 6.10 (Measure on Semi-ring)**.** Let $\mathcal{S}$ be a semi-ring. A set function

$$m : \mathcal{S} \to [0, \infty]$$

is a measure if:

1.
$$m(\varnothing) = 0.$$

2. **Finite Additivity:** If
$$A = \bigsqcup_{i=1}^{n} A_i$$
with $A, A_i \in \mathcal{S}$, then
$$m(A) = \sum_{i=1}^{n} m(A_i).$$

3. **Countable Additivity (σ-additivity):** If

$$A = \bigsqcup_{i=1}^{\infty} A_i$$

with $A, A_i \in \mathcal{S}$, then

$$m(A) = \sum_{i=1}^{\infty} m(A_i).$$

Example 6.11. On the semi-ring of intervals

$$\mathcal{S} = \{[a, b)\},$$

the function

$$m([a, b)) = b - a$$

is a countably additive measure (standard length).

Proposition 6.12 (Extension to Ring)**.** *Let m be a finitely additive measure on a semi-ring $\mathcal{S}$. There exists a unique extension of m to the generated ring $\mathcal{R}(\mathcal{S})$ which is also finitely additive. If m is countably additive on $\mathcal{S}$, the extension is countably additive on $\mathcal{R}(\mathcal{S})$.*

Additional Proof. For a disjoint representation

$$A = \bigsqcup_{i=1}^{n} A_i, \qquad A_i \in \mathcal{S},$$

define

$$m(A) = \sum_{i=1}^{n} m(A_i).$$

The key point is that this value is independent of the chosen decomposition.

Indeed, suppose also

$$A = \bigsqcup_{j=1}^{m} B_j.$$

Then

$$A_i = \bigsqcup_{j=1}^{m} (A_i \cap B_j), \qquad B_j = \bigsqcup_{i=1}^{n} (A_i \cap B_j),$$

after decomposing the intersections into semi-ring pieces when necessary. Finite additivity gives

$$\sum_{i=1}^{n} m(A_i) = \sum_{i,j} m(A_i \cap B_j) = \sum_{j=1}^{m} m(B_j).$$

Hence the extension is well-defined. Finite additivity follows immediately by combining disjoint decompositions. The same refinement argument together with countable additivity proves the last statement. □

6.3 Outer Measure and Carathéodory's Theorem

We now wish to extend the measure from the ring $\mathcal{R}(\mathcal{S})$ to a σ-algebra.

Definition 6.13 (Outer Measure)**.** Let m be a measure on a semi-ring $\mathcal{S}$. For any subset $E \subset \Omega$, the Lebesgue Outer Measure $\mu^*(E)$ is defined as:

$$\mu^*(E) = \inf \left\{ \sum_{j=1}^{\infty} m(A_j) : E \subset \bigcup_{j=1}^{\infty} A_j,\ A_j \in \mathcal{S} \right\}.$$

Remark 6.14. The idea is to cover E by countably many simple measurable pieces and take the infimum of their total measure.

Proposition 6.15 (Properties of μ^*)**.** *1.*

$$\mu^*(\varnothing) = 0.$$

2. ***Monotonicity:***

$$A \subset B \Longrightarrow \mu^*(A) \leq \mu^*(B).$$

3. ***Countable Subadditivity:*** *For any sequence of sets* $\{E_j\}$*,*

$$\mu^* \left(\bigcup_{j=1}^{\infty} E_j \right) \leq \sum_{j=1}^{\infty} \mu^*(E_j).$$

Additional Proof. The first property follows because the empty family gives a cover of $\varnothing$ of total measure 0.

For monotonicity, if $A \subset B$, then every countable cover of B is also a cover of A. Therefore the infimum defining $\mu^*(A)$ is taken over a larger collection of covers, so

$$\mu^*(A) \leq \mu^*(B).$$

For countable subadditivity, fix $\varepsilon > 0$. For every j, choose a cover

$$E_j \subset \bigcup_{k=1}^{\infty} A_{jk}$$

such that

$$\sum_{k=1}^{\infty} m(A_{jk}) < \mu^*(E_j) + \frac{\varepsilon}{2^j}.$$

Then the collection $\{A_{jk}\}_{j,k}$ covers $\bigcup_j E_j$, so

$$\mu^* \left(\bigcup_{j=1}^{\infty} E_j \right) \leq \sum_{j=1}^{\infty} \sum_{k=1}^{\infty} m(A_{jk}) < \sum_{j=1}^{\infty} \mu^*(E_j) + \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary,

$$\mu^* \left(\bigcup_{j=1}^{\infty} E_j \right) \leq \sum_{j=1}^{\infty} \mu^*(E_j).$$

□

The outer measure is defined for all subsets, but it is not necessarily additive. We identify the "good" sets (measurable sets) using Carathéodory's condition.

Definition 6.16 (Carathéodory Measurability)**.** A subset $E \subset \Omega$ is called μ^*-measurable if for every set $A \subset \Omega$ (test set):

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \setminus E).$$

Intuitively, E splits every set A additively.

Theorem 6.17 (Carathéodory's Extension Theorem)**.** *Let $\mathcal{M}$ be the collection of all μ^*-measurable sets.*

1. *$\mathcal{M}$ is a σ-algebra containing $\mathcal{S}$.*
2. *The restriction*
$$\mu = \mu^*|_{\mathcal{M}}$$
is a countably additive measure (a complete measure).
3. *For $A \in \mathcal{S}$,*
$$\mu(A) = m(A)$$
(Consistent extension).

Proof Idea.

1. Algebra: $E \in \mathcal{M} \iff E^c \in \mathcal{M}$ by symmetry of definition. Intersection is shown using subadditivity.
2. σ-algebra: Disjoint unions work by induction and subadditivity.
3. Completeness: If $\mu^*(E) = 0$, then for any A, $\mu^*(A \cap E) = 0$ and $\mu^*(A \setminus E) \leq \mu^*(A)$, so equality holds.

Proof. We give the details of the main assertions.

1. $\mathcal{M}$ is an algebra.

First, $\Omega \in \mathcal{M}$ because for every test set A,

$$\mu^*(A) = \mu^*(A \cap \Omega) + \mu^*(A \setminus \Omega) = \mu^*(A) + 0.$$

If $E \in \mathcal{M}$, then for every A,

$$\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c).$$

The condition is symmetric in E and E^c, hence

$$E^c \in \mathcal{M}.$$

Now let $E_1, E_2 \in \mathcal{M}$. For any test set A, measurability of E_1 gives

$$\mu^*(A) = \mu^*(A \cap E_1) + \mu^*(A \cap E_1^c).$$

Apply measurability of E_2 to each term:

$$\mu^*(A \cap E_1) = \mu^*(A \cap E_1 \cap E_2) + \mu^*(A \cap E_1 \cap E_2^c),$$

and

$$\mu^*(A \cap E_1^c) = \mu^*(A \cap E_1^c \cap E_2) + \mu^*(A \cap E_1^c \cap E_2^c).$$

Thus

$$\begin{aligned}\mu^*(A) = {} & \mu^*(A \cap E_1 \cap E_2) + \mu^*(A \cap E_1 \cap E_2^c) \\ & + \mu^*(A \cap E_1^c \cap E_2) + \mu^*(A \cap E_1^c \cap E_2^c).\end{aligned}$$

By subadditivity,

$$\mu^*(A \cap (E_1 \cup E_2)) \leq \mu^*(A \cap E_1 \cap E_2) + \mu^*(A \cap E_1 \cap E_2^c) + \mu^*(A \cap E_1^c \cap E_2),$$

and therefore

$$\mu^*(A) \geq \mu^*(A \cap (E_1 \cup E_2)) + \mu^*(A \cap (E_1 \cup E_2)^c).$$

The reverse inequality follows from countable subadditivity, so equality holds. Hence

$$E_1 \cup E_2 \in \mathcal{M}.$$

Therefore $\mathcal{M}$ is an algebra.

2. μ^* is finitely additive on disjoint measurable sets.

Let

$$E_1, E_2 \in \mathcal{M}, \qquad E_1 \cap E_2 = \varnothing.$$

Take the test set

$$A = E_1 \cup E_2.$$

Since E_1 is measurable,

$$\mu^*(E_1 \cup E_2) = \mu^*((E_1 \cup E_2) \cap E_1) + \mu^*((E_1 \cup E_2) \setminus E_1),$$

hence

$$\mu^*(E_1 \cup E_2) = \mu^*(E_1) + \mu^*(E_2).$$

3. $\mathcal{M}$ is a σ-algebra.

Let $\{E_j\}_{j=1}^{\infty} \subset \mathcal{M}$ be pairwise disjoint, and define

$$S_n = \bigcup_{j=1}^{n} E_j, \qquad S = \bigcup_{j=1}^{\infty} E_j.$$

Since $\mathcal{M}$ is an algebra,

$$S_n \in \mathcal{M}.$$

Thus for every test set A,

$$\mu^*(A) = \mu^*(A \cap S_n) + \mu^*(A \cap S_n^c).$$

By finite additivity,

$$\mu^*(A \cap S_n) = \sum_{j=1}^{n} \mu^*(A \cap E_j).$$

Therefore

$$\mu^*(A) = \sum_{j=1}^{n} \mu^*(A \cap E_j) + \mu^*(A \cap S_n^c).$$

Since

$$A \cap S^c \subset A \cap S_n^c,$$

monotonicity gives

$$\mu^*(A) \geq \sum_{j=1}^{n} \mu^*(A \cap E_j) + \mu^*(A \cap S^c).$$

Letting $n \to \infty$,

$$\mu^*(A) \geq \sum_{j=1}^{\infty} \mu^*(A \cap E_j) + \mu^*(A \cap S^c).$$

By countable subadditivity,

$$\mu^*(A \cap S) \leq \sum_{j=1}^{\infty} \mu^*(A \cap E_j),$$

hence

$$\mu^*(A) \geq \mu^*(A \cap S) + \mu^*(A \cap S^c).$$

The reverse inequality follows from subadditivity:

$$\mu^*(A) \leq \mu^*(A \cap S) + \mu^*(A \cap S^c).$$

Thus

$$\mu^*(A) = \mu^*(A \cap S) + \mu^*(A \cap S^c),$$

so

$$S \in \mathcal{M}.$$

Hence $\mathcal{M}$ is closed under countable disjoint unions. Since it is already an algebra, it is a σ-algebra.

4. μ^* is σ-additive on $\mathcal{M}$.

Let

$$S = \bigsqcup_{j=1}^{\infty} E_j, \qquad E_j \in \mathcal{M}.$$

Using the preceding finite-additivity argument,

$$\mu^*(S) \geq \sum_{j=1}^{n} \mu^*(E_j)$$

for every n. Therefore

$$\mu^*(S) \geq \sum_{j=1}^{\infty} \mu^*(E_j).$$

On the other hand, countable subadditivity gives

$$\mu^*(S) \leq \sum_{j=1}^{\infty} \mu^*(E_j).$$

Hence

$$\mu^*(S) = \sum_{j=1}^{\infty} \mu^*(E_j).$$

Finally, if $\mu^*(E) = 0$, then every subset $N \subset E$ also satisfies

$$\mu^*(N) = 0$$

by monotonicity. Such a set satisfies Carathéodory's condition, so $N \in \mathcal{M}$. Therefore the restricted measure is complete. □

6.4 Lebesgue Measure on $\mathbb{R}^n$

Definition 6.18 (Borel and Lebesgue σ-algebras)**.**

- Let $\mathcal{S}$ be the semi-ring of cells in $\mathbb{R}^n$. The σ-algebra generated by $\mathcal{S}$ (or equivalently by open sets) is the Borel σ-algebra

$$\mathcal{B}(\mathbb{R}^n).$$

- The σ-algebra $\mathcal{M}$ obtained via Carathéodory's theorem is the Lebesgue σ-algebra

$$\mathcal{L}(\mathbb{R}^n).$$

- We have strict inclusion:

$$\mathcal{B}(\mathbb{R}^n) \subsetneq \mathcal{L}(\mathbb{R}^n).$$

Remark 6.19. Borel sets are generated from open sets by countable unions, countable intersections, and complements. Lebesgue measurable sets form a larger class because the Lebesgue σ-algebra also contains all subsets of sets of Lebesgue measure zero.

Proposition 6.20 (Regularity)**.** *Lebesgue measure is regular. For any measurable set E and $\varepsilon > 0$:*

- ***Outer regularity:*** *$\exists$ open $G \supset E$ such that*

$$\mu(G \setminus E) < \varepsilon.$$

- ***Inner regularity:*** *$\exists$ compact $K \subset E$ such that*

$$\mu(E \setminus K) < \varepsilon$$

(if $\mu(E) < \infty$).

6.5 Non-measurable Sets

Does $\mathcal{M}$ contain all subsets of $\mathbb{R}^n$? The answer is no, assuming the Axiom of Choice.

Theorem 6.21 (Existence of Non-measurable Sets)**.** *There exists a subset of $\mathbb{R}$ that is not Lebesgue measurable.*

Construction (Vitali Set). Define an equivalence relation on $[0, 1]$:

$$x \sim y \iff x - y \in \mathbb{Q}.$$

This partitions $[0, 1]$ into disjoint equivalence classes $\{E_\alpha\}$. By the Axiom of Choice, form a set V by selecting exactly one representative from each equivalence class.

Let $\{r_k\}_{k=1}^{\infty}$ be an enumeration of

$$\mathbb{Q} \cap [-1, 1].$$

Consider the translations:

$$V_k = V + r_k = \{v + r_k : v \in V\}.$$

Properties:

1. The sets V_k are disjoint. (If
$$v + r_i = v' + r_j,$$
then
$$v - v' \in \mathbb{Q},$$
so $v \sim v'$, so $v = v'$ and $i = j$.)
2. $$[0,1] \subset \bigcup_{k=1}^{\infty} V_k \subset [-1,2].$$
(For any $x \in [0,1]$, $x \sim v$ for some $v \in V$, so
$$x - v \in \mathbb{Q} \cap [-1,1].$$
)

Assume V is measurable. Since Lebesgue measure is translation invariant,
$$\mu(V_k) = \mu(V).$$
By σ-additivity:
$$\mu\left(\bigcup_{k=1}^{\infty} V_k\right) = \sum_{k=1}^{\infty} \mu(V).$$
However, using monotonicity on property 2:
$$1 \leq \sum_{k=1}^{\infty} \mu(V) \leq 3.$$

- If $\mu(V) = 0$, the sum is 0, contradicting
$$1 \leq 0.$$
- If $\mu(V) > 0$, the sum is infinite, contradicting
$$\infty \leq 3.$$

Thus, V cannot be measurable.

Remark 6.22 (Banach-Tarski Paradox). In $\mathbb{R}^3$ and higher, one can decompose the unit ball into finite pieces and reassemble them (using rigid motions) to form two identical balls. This relies heavily on non-measurable sets.

Chapter 7

The Cantor Set and Cantor Function

7 The Cantor Set and Cantor Function

This chapter introduces one of the most important counterexamples in analysis. The Cantor set is a "small" set (measure zero, nowhere dense) that is topologically "large" (uncountable, perfect). The associated Cantor function demonstrates that a continuous, monotone function can grow from 0 to 1 despite having a derivative of zero almost everywhere.

7.1 Construction of the Cantor Ternary Set

We construct the Cantor set $C \subset [0,1]$ by iteratively removing "middle third" open intervals.

Definition 7.1 (Construction)**.** Let

$$C_0 = [0,1].$$

1. **Step 1:** Remove the open middle third interval
$$\left(\frac{1}{3}, \frac{2}{3}\right)$$
from C_0.
$$C_1 = \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right].$$
C_1 consists of 2^1 closed intervals, each of length 3^{-1}.
2. **Step 2:** Remove the open middle third of each of the 2 intervals in C_1.
$$C_2 = \left[0, \frac{1}{9}\right] \cup \left[\frac{2}{9}, \frac{1}{3}\right] \cup \left[\frac{2}{3}, \frac{7}{9}\right] \cup \left[\frac{8}{9}, 1\right].$$
C_2 consists of 2^2 closed intervals, each of length 3^{-2}.
3. **Step n:** In general, C_n is obtained from C_{n-1} by removing the open middle third of length 3^{-n} from each of the 2^{n-1} intervals comprising C_{n-1}. C_n consists of 2^n disjoint closed intervals of length 3^{-n}.

The Cantor Ternary Set is defined as the intersection of all these sets:

$$C := \bigcap_{n=0}^{\infty} C_n.$$

7.2 Properties of the Cantor Set

Theorem 7.2. *The Cantor set C has the following properties:*

1. *C is compact.*
2. *C has Lebesgue measure zero ($\mu(C) = 0$).*
3. *C is nowhere dense.*
4. *C is uncountable (has cardinality $\mathfrak{c}$).*
5. *C is a perfect set (closed and has no isolated points).*

Proof. **1. Compactness:** C is an intersection of closed sets in $[0,1]$, so it is closed. Since it is a subset of $[0,1]$, it is bounded. By Heine-Borel, C is compact.

2. Measure Zero: Let us calculate the total length of the removed open intervals

$$U = [0,1] \setminus C.$$

- Step 1: Removed 1 interval of length $1/3$.
- Step 2: Removed 2 intervals of length $1/9$.
- Step k: Removed 2^{k-1} intervals of length $1/3^k$.

Total length removed:

$$\mu(U) = \sum_{k=1}^{\infty} \frac{2^{k-1}}{3^k} = \frac{1}{3} \sum_{k=0}^{\infty} \left(\frac{2}{3}\right)^k = \frac{1}{3} \cdot \frac{1}{1 - 2/3} = \frac{1}{3} \cdot 3 = 1.$$

Thus,

$$\mu(C) = \mu([0,1]) - \mu(U) = 1 - 1 = 0.$$

3. Nowhere Dense: C is closed. Does it contain any open interval? If C contained an interval (a, b), then

$$\mu(C) \geq b - a > 0,$$

which contradicts $\mu(C) = 0$. Since C contains no open sets, its interior is empty. Thus C is nowhere dense.

4. Uncountability (Ternary Representation): Any $x \in [0,1]$ has a ternary expansion

$$x = \sum_{k=1}^{\infty} \frac{a_k}{3^k}$$

with digits

$$a_k \in \{0, 1, 2\}.$$

- Removing

$$\left(\frac{1}{3}, \frac{2}{3}\right)$$

removes numbers with $a_1 = 1$.

- Removing middle thirds at step n removes numbers with $a_n = 1$.

Elements in C are exactly those x which admit a ternary expansion consisting only of digits 0 and 2. Consider the mapping

$$f : C \to [0,1]$$

defined by converting the digits $0 \to 0$ and $2 \to 1$ to form a binary expansion:

$$x = \sum_{k=1}^{\infty} \frac{2b_k}{3^k} \in C \xmapsto{f} y = \sum_{k=1}^{\infty} \frac{b_k}{2^k} \in [0,1], \qquad b_k \in \{0, 1\}.$$

This mapping is surjective onto $[0,1]$. Thus

$$|C| \geq |[0,1]| = \mathfrak{c}.$$

Since

$$C \subset \mathbb{R},$$

we have

$$|C| \leq \mathfrak{c}.$$

Hence

$$|C| = \mathfrak{c}.$$

5. Perfectness: We already know that C is closed. It remains to show that C has no isolated points.

Let $x \in C$. For every n, the point x belongs to one of the closed intervals comprising C_n, and that interval has length 3^{-n}. Inside that interval there is another point

$$y_n \in C, \qquad y_n \neq x.$$

Moreover,

$$|x - y_n| \leq 3^{-n} \to 0.$$

Hence

$$y_n \to x.$$

Therefore every point of C is an accumulation point of C, so C has no isolated points. Thus C is perfect. □

Remark 7.3. A perfect set is a closed set with no isolated points. Equivalently, every point of the set is an accumulation point of the set.

7.3 The Cantor-Lebesgue Function (Devil's Staircase)

We construct a continuous function

$$K : [0,1] \to [0,1]$$

that is locally constant on the complement of C.

Definition 7.4 (Construction)**.** Let

$$U = [0,1] \setminus C = \bigsqcup_{k=1}^{\infty} I_k$$

be the countable union of disjoint removed open intervals.

1. On the interval removed in Step 1,

$$I_1 = \left(\frac{1}{3}, \frac{2}{3}\right),$$

   define

$$K(x) = \frac{1}{2}.$$

2. On the intervals removed in Step 2,

$$I_{2,1} = \left(\frac{1}{9}, \frac{2}{9}\right), \qquad I_{2,2} = \left(\frac{7}{9}, \frac{8}{9}\right),$$

   define

$$K(x) = \frac{1}{4}$$

   and

$$K(x) = \frac{3}{4}$$

   respectively.
3. In general, on the 2^{n-1} intervals removed at Step n, define $K(x)$ to be the midpoint of the values taken on the adjacent intervals from previous steps.

This defines K on the dense set U. We extend it to C (and thus $[0,1]$) by continuity:

$$K(x) = \sup\{K(t) : t \in U,\ t < x\}.$$

Definition 7.5 (Analytical Definition)**.** For $x \in C$ with ternary expansion

$$x = \sum_{k=1}^{\infty} \frac{a_k}{3^k}$$

where

$$a_k \in \{0, 2\},$$
$$K(x) = \sum_{k=1}^{\infty} \frac{a_k/2}{2^k}.$$

This maps the "ternary Cantor set" to the "binary interval $[0,1]$".

Proposition 7.6 (Properties of K)**.** *1.* ***Continuity:*** *K is continuous on $[0,1]$. It does not jump because the range values are dense in $[0,1]$.*

2. ***Monotonicity:*** *K is non-decreasing. $K(0) = 0$, $K(1) = 1$.*

3. ***Derivative:*** *Since K is constant on every interval in*

$$U = [0,1] \setminus C,$$

and $\mu(U) = 1$:

$$K'(x) = 0$$

for almost every $x \in [0,1]$.

Additional Details. For monotonicity, take $x < y$. Their ternary expansions agree up to some first digit where they differ. On the Cantor set, the only possible digits are 0 and 2, and replacing

$$0 \mapsto 0, \qquad 2 \mapsto 1$$

preserves lexicographic order. Hence

$$K(x) \leq K(y).$$

On each removed interval, K is constant, so the monotonicity extends to all of $[0,1]$.

For continuity, fix $x \in [0,1]$. At stage n, x belongs either to a deleted interval, where K is locally constant, or to one of the remaining intervals of length 3^{-n}. Across such an interval the total possible change of K is at most

$$2^{-n}.$$

Thus, as $n \to \infty$, both the horizontal scale 3^{-n} and the corresponding variation 2^{-n} tend to zero. Hence K is continuous.

Finally,

$$U = [0,1] \setminus C$$

is an open set of measure 1, and K is locally constant on every connected component of U. Therefore

$$K'(x) = 0$$

for every $x \in U$, and since

$$\mu(C) = 0,$$

we conclude

$$K'(x) = 0$$

for almost every $x \in [0,1]$. □

Remark 7.7 (Singular Function). The Cantor function is a prime example of a singular function: it is continuous and non-constant, yet its derivative vanishes almost everywhere. This shows that the Fundamental Theorem of Calculus (Newton-Leibniz) fails for general continuous functions:

$$\int_0^1 K'(x)\,dx = \int_0^1 0\,dx = 0 \neq K(1) - K(0) = 1.$$

This motivates the definition of Absolute Continuity (covered in Part IV).

Exercise 7.8. Consider the function

$$\varphi(x) = x + K(x).$$

- φ is strictly increasing, continuous bijection from $[0,1]$ to $[0,2]$.
- φ maps the measure-zero set C to a set of positive measure (since

$$\mu(\varphi(C)) = \mu([0,2]) - \mu(\varphi(U)) = 2 - 1 = 1).$$

- This map φ is often used to construct counterexamples (e.g., showing that the image of a measurable set under a continuous map need not be measurable).

Chapter 8

Measurable Functions

8 Measurable Functions

8.1 Definitions and Algebraic Properties

Definition 8.1. Let $(\Omega, \mathcal{A}, \mu)$ be a measurable space. A function

$$f : \Omega \to \mathbb{R}$$

is called measurable if the preimage of every Borel set in $\mathbb{R}$ is a measurable set in Ω. That is,

$$\forall B \in \mathcal{B}(\mathbb{R}), \qquad f^{-1}(B) = \{\omega \in \Omega : f(\omega) \in B\} \in \mathcal{A}.$$

Proposition 8.2 (Equivalent Definitions)**.** *The condition for measurability can be checked on generators of the Borel σ-algebra. The following are equivalent:*

1. *f is measurable.*
2. *$\forall c \in \mathbb{R}$,*
$$\{x : f(x) > c\} \in \mathcal{A}.$$
3. *$\forall c \in \mathbb{R}$,*
$$\{x : f(x) \geq c\} \in \mathcal{A}.$$
4. *$\forall c \in \mathbb{R}$,*
$$\{x : f(x) < c\} \in \mathcal{A}.$$

Additional Proof. Clearly, if f is measurable, then all the sets in (2)–(4) are measurable because

$$(c, \infty), \qquad [c, \infty), \qquad (-\infty, c)$$

are Borel sets.

Conversely, suppose for example that

$$\{x : f(x) > c\} \in \mathcal{A}$$

for every $c \in \mathbb{R}$. Then

$$\{x : f(x) \leq c\} = \{x : f(x) > c\}^c \in \mathcal{A}.$$

Also,

$$\{x : f(x) < c\} = \bigcup_{n=1}^{\infty} \left\{x : f(x) \leq c - \frac{1}{n}\right\} \in \mathcal{A}.$$

Thus the preimage of every open interval is measurable. Since open intervals generate the Borel σ-algebra, f is measurable. □

Remark 8.3. In the definition, one can replace "Borel set" with open sets or closed sets. The class of measurable functions is the smallest class containing characteristic functions of measurable sets and closed under pointwise limits.

Theorem 8.4 (Algebraic Properties)**.** *Let f, g be measurable functions and $c \in \mathbb{R}$. Then the following functions are measurable:*

1. *cf, $f + c$, f^2.*
2. *$f + g$, $f \cdot g$.*
3. *$\max(f, g)$, $\min(f, g)$, $|f|$.*

4. If $f(x) \neq 0$ everywhere, then $1/f$ is measurable.

Proof for Sum. Observe that

$$\{f + g < c\} = \bigcup_{r \in \mathbb{Q}} \left(\{f < r\} \cap \{g < c - r\}\right).$$

Since $\mathbb{Q}$ is countable and intersections/unions of measurable sets are measurable, the result follows.

Additional Details. The remaining assertions follow from similar identities or composition with continuous functions.

For example,

$$f^2 = (f)^2$$

is measurable because $x \mapsto x^2$ is continuous. Likewise,

$$|f|$$

is measurable.

Also,

$$\max(f, g) = \frac{f + g + |f - g|}{2},$$

and

$$\min(f, g) = \frac{f + g - |f - g|}{2}.$$

Hence both are measurable.

For the product, use

$$fg = \frac{1}{4}\left((f + g)^2 - (f - g)^2\right).$$

Finally, if f never vanishes, then

$$\frac{1}{f}$$

is measurable because the map

$$x \mapsto \frac{1}{x}$$

is continuous on $\mathbb{R} \setminus \{0\}$. □

Theorem 8.5 (Limits)**.** *Let $\{f_n\}$ be a sequence of measurable functions. Then the following are measurable:*

$$\sup_n f_n, \qquad \inf_n f_n, \qquad \limsup_{n\to\infty} f_n, \qquad \liminf_{n\to\infty} f_n.$$

Consequently, if

$$f(x) = \lim_{n\to\infty} f_n(x)$$

exists pointwise, then f is measurable.

Proof. For the supremum, for every $c \in \mathbb{R}$,

$$\left\{x : \sup_n f_n(x) > c\right\} = \bigcup_{n=1}^{\infty} \{x : f_n(x) > c\}.$$

The right-hand side is measurable, so $\sup_n f_n$ is measurable.

Similarly,

$$\inf_n f_n = -\sup_n(-f_n),$$

so $\inf_n f_n$ is measurable.

Now

$$\limsup_{n\to\infty} f_n = \inf_{n\geq 1} \sup_{k\geq n} f_k,$$

and

$$\liminf_{n\to\infty} f_n = \sup_{n\geq 1} \inf_{k\geq n} f_k.$$

Therefore both are measurable.

If $f_n(x) \to f(x)$ pointwise, then

$$f = \limsup_{n\to\infty} f_n = \liminf_{n\to\infty} f_n,$$

so f is measurable. □

8.2 Convergence of Measurable Functions

We distinguish several modes of convergence for functions on a measure space $(\Omega, \mathcal{A}, \mu)$.

Definition 8.6 (Modes of Convergence)**.** Let f_n, f be measurable.

1. **Pointwise Convergence:**
$$f_n(x) \to f(x)$$
for all $x \in \Omega$.
2. **Uniform Convergence:**
$$\sup_{x\in\Omega} |f_n(x) - f(x)| \to 0$$
as $n \to \infty$.
3. **Almost Everywhere (a.e.) Convergence:**
$$f_n(x) \to f(x)$$
for all $x \in \Omega \setminus E$, where
$$\mu(E) = 0.$$
Notation:
$$f_n \xrightarrow{\text{a.e.}} f.$$

Remark 8.7. Pointwise convergence requires convergence at every point. Almost everywhere convergence allows failure on a set of measure zero. Uniform convergence is stronger because the rate of convergence is controlled simultaneously over the entire domain:

$$\max_x |f_n(x) - f(x)| \to 0$$

when the maximum exists.

Definition 8.8 (Essential Supremum)**.** For a measurable function f,

$$\|f\|_\infty = \operatorname{ess\,sup} |f| = \inf\{C \geq 0 : |f(x)| \leq C \text{ a.e.}\}.$$

Remark 8.9. The essential supremum ignores sets of measure zero. Thus changing f on a null set does not change $\|f\|_\infty$.

8.3 Egorov's Theorem

Uniform convergence is very strong. Egorov's theorem states that on a finite measure space, pointwise convergence is "almost" uniform.

Theorem 8.10 (Egorov)**.** *Let $(\Omega, \mathcal{A}, \mu)$ be a measure space with*

$$\mu(\Omega) < \infty.$$

Let $\{f_n\}$ be a sequence of measurable functions converging to f a.e. Then, for every $\varepsilon > 0$, there exists a measurable set $E \subset \Omega$ such that:

1. $$\mu(E) < \varepsilon.$$
2. *f_n converges to f uniformly on $\Omega \setminus E$.*

Proof. Without loss of generality, assume

$$f_n \to f$$

everywhere (redefine on null set). For $k, n \in \mathbb{N}$, define

$$E_n^k = \bigcup_{m=n}^{\infty} \left\{ x : |f_m(x) - f(x)| \geq \frac{1}{k} \right\}.$$

For a fixed k, the sequence sets $\{E_n^k\}_{n=1}^{\infty}$ is decreasing:

$$E_{n+1}^k \subset E_n^k.$$

Since $f_m(x) \to f(x)$, for any x, there exists N such that for all $m \geq N$,

$$|f_m(x) - f(x)| < \frac{1}{k}.$$

Thus

$$\bigcap_{n=1}^{\infty} E_n^k = \varnothing.$$

Since

$$\mu(\Omega) < \infty,$$

by continuity of measure from above,

$$\lim_{n \to \infty} \mu(E_n^k) = 0.$$

Thus, for the given ε, and each k, we can choose an integer N_k such that

$$\mu(E_{N_k}^k) < \frac{\varepsilon}{2^k}.$$

Define

$$E = \bigcup_{k=1}^{\infty} E_{N_k}^k.$$

Then

$$\mu(E) \leq \sum_{k=1}^{\infty} \mu(E_{N_k}^k) < \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon.$$

On the complement

$$F = \Omega \setminus E,$$

we have

$$x \notin E_{N_k}^k$$

for any k. This means

$$\forall k,\ \forall m \geq N_k, \qquad |f_m(x) - f(x)| < \frac{1}{k}.$$

Since N_k depends only on k (not on x), the convergence is uniform on F. □

Remark 8.11. The construction removes, for each accuracy level $1/k$, the set on which the tail

$$f_n, f_{n+1}, \dots$$

has not yet entered the $1/k$-neighborhood of f. The exceptional sets are chosen so that their total measure is less than ε.

Remark 8.12. The assumption

$$\mu(\Omega) < \infty$$

is necessary. Consider

$$f_n(x) = \chi_{[n,n+1]}(x)$$

on $\mathbb{R}$. $f_n \to 0$ pointwise everywhere, but convergence is not uniform on any set of finite measure complement (specifically, on any unbounded set).

8.4 Convergence in Measure

Definition 8.13. Let sequence $\{f_n\}$ and f be finite a.e. We say f_n converges in measure to f (denoted

$$f_n \xrightarrow{\mu} f$$

) if for every $\sigma > 0$:

$$\lim_{n\to\infty} \mu\left(\{x \in \Omega : |f_n(x) - f(x)| \geq \sigma\}\right) = 0.$$

Proposition 8.14. *1. If $\mu(\Omega) < \infty$, then*

$$f_n \xrightarrow{\text{a.e.}} f \Longrightarrow f_n \xrightarrow{\mu} f.$$

(Follows from Egorov).

2.

$$f_n \to f \ \textit{uniformly} \Longrightarrow f_n \xrightarrow{\mu} f$$

(on finite measure sets).

Additional Proof. For (1), let $\sigma > 0$ and $\varepsilon > 0$. By Egorov's theorem, there exists

$$E \subset \Omega$$

such that

$$\mu(E) < \varepsilon$$

and $f_n \to f$ uniformly on $\Omega \setminus E$. Hence for sufficiently large n,

$$|f_n(x) - f(x)| < \sigma$$

for every $x \in \Omega \setminus E$. Therefore

$$\{x : |f_n(x) - f(x)| \geq \sigma\} \subset E,$$

so

$$\mu\{x : |f_n - f| \geq \sigma\} \leq \mu(E) < \varepsilon.$$

Thus

$$f_n \xrightarrow{\mu} f.$$

For (2), if convergence is uniform, then for sufficiently large n,

$$|f_n(x) - f(x)| < \sigma$$

for every x. Hence

$$\{x : |f_n(x) - f(x)| \geq \sigma\} = \varnothing,$$

so its measure is zero. □

Example 8.15 (Typewriter Sequence / Moving Bump)**.** Consider f_n on $[0, 1]$ defined by characteristic functions of intervals:

$$[0, 1],\ [0, 1/2],\ [1/2, 1],\ [0, 1/4],\ [1/4, 1/2], \ldots$$

This sequence converges to 0 in measure (the width of the support goes to 0). However, for every $x \in [0, 1]$, $f_n(x) = 1$ for infinitely many n, so $f_n(x)$ does not converge to 0. Thus

$$f_n \not\to 0 \quad \text{a.e.}$$

Remark 8.16. This example shows that convergence in measure does **not** imply almost everywhere convergence for the full sequence.

Theorem 8.17 (Riesz)**.** *If*

$$f_n \xrightarrow{\mu} f,$$

then there exists a subsequence $\{f_{n_k}\}$ *such that*

$$f_{n_k} \xrightarrow{\text{a.e.}} f.$$

Sketch. Since $f_n \to f$ in measure, for each k, we can choose n_k such that

$$\mu\left(\{|f_{n_k} - f| \geq 2^{-k}\}\right) < 2^{-k}.$$

Let

$$A_k = \{|f_{n_k} - f| \geq 2^{-k}\}.$$

Then

$$\sum_{k=1}^{\infty} \mu(A_k) < \infty.$$

By Borel-Cantelli Lemma, the set of x belonging to infinitely many A_k has measure zero. For x not in this set,

$$|f_{n_k}(x) - f(x)| < 2^{-k}$$

for large k, implying convergence.

Remark 8.18. The Borel–Cantelli step says that, outside a null set, each point belongs to only finitely many of the bad sets A_k. Therefore, for almost every x, eventually

$$x \notin A_k,$$

which is exactly

$$|f_{n_k}(x) - f(x)| < 2^{-k}.$$

8.5 Lusin's Theorem

Littlewood's Second Principle: "Every measurable function is nearly continuous."

Theorem 8.19 (Lusin)**.** *Let f be a measurable function on $\mathbb{R}$ (or a domain $\Omega \subset \mathbb{R}^n$ with finite measure). For every $\varepsilon > 0$, there exists a continuous function*

$$g : \mathbb{R} \to \mathbb{R}$$

(or $g \in C_c(\mathbb{R})$) such that

$$\mu(\{x : f(x) \neq g(x)\}) < \varepsilon.$$

More precisely, there exists a closed set

$$F \subset \Omega$$

with

$$\mu(\Omega \setminus F) < \varepsilon$$

such that the restriction

$$f|_F$$

is continuous.

Remark 8.20. Note that $f|_F$ being continuous does not imply f is continuous at points of F as a function on Ω. It means continuity relative to the subspace topology of F. However, by Tietze Extension Theorem, $f|_F$ can be extended to a continuous function g on all of $\mathbb{R}$.

Idea.

1. The theorem holds for simple functions (characteristic functions of intervals are nearly continuous; approximate rectangular pulses by trapezoids).
2. Any measurable function is a limit of simple functions.
3. Use Egorov's theorem to make the convergence uniform on a large set. Uniform limit of continuous functions is continuous.

Remark 8.21. Lusin's theorem should be understood as saying that a measurable function may be highly irregular globally, but after removing a set of arbitrarily small measure, its restriction behaves like a continuous function.

Chapter 9

The Lebesgue Integral

9 The Lebesgue Integral

We now define the Lebesgue integral. The approach differs from Riemann integration: instead of partitioning the domain (the x-axis), we partition the range (the y-axis). This allows us to integrate a much wider class of functions (e.g., unbounded functions, functions on fractal sets) and provides better limit theorems.

9.1 The Integral of Simple Functions

Definition 9.1 (Simple Function)**.** A function $\varphi : \Omega \to \mathbb{R}$ is called simple if it takes only a finite number of distinct values. If φ assumes distinct non-zero values $c_1, \ldots, c_n$ on disjoint sets $E_1, \ldots, E_n$, we write the canonical representation:

$$\varphi(x) = \sum_{j=1}^{n} c_j \chi_{E_j}(x),$$

where χ_E is the characteristic function ($\chi_E(x) = 1$ if $x \in E$, 0 otherwise).

Remark 9.2. Some texts (and occasionally these notes) allow simple functions to take countably many values. If so, absolute convergence of the defining series is required. However, for the standard construction, we assume finite values first.

Lemma 9.3. *The sum and product of simple measurable functions are simple measurable functions.*

Additional Proof. Suppose

$$\varphi = \sum_{i=1}^{m} a_i \chi_{A_i}, \qquad \psi = \sum_{j=1}^{n} b_j \chi_{B_j}.$$

Consider the common refinement

$$C_{ij} = A_i \cap B_j.$$

On each C_{ij}, both φ and ψ are constant. Hence

$$\varphi + \psi$$

takes only the finitely many possible values $a_i + b_j$, and

$$\varphi\psi$$

takes only the finitely many possible values $a_i b_j$. Since the sets C_{ij} are measurable, both functions are simple measurable. □

Definition 9.4 (Integral of a Simple Function)**.** Let

$$\varphi = \sum_{j=1}^{n} c_j \chi_{E_j}$$

be a simple measurable function on $(\Omega, \mathcal{A}, \mu)$. For any measurable subset $E \in \mathcal{A}$, we define the integral of φ over E as:

$$\int_E \varphi \, d\mu = \sum_{j=1}^{n} c_j \mu(E \cap E_j).$$

We adopt the convention

$$0 \cdot \infty = 0.$$

Proposition 9.5 (Independence of Representation)**.** *The definition of the integral does not depend on the representation of φ. If*

$$\varphi = \sum_{i=1}^{n} a_i \chi_{A_i} = \sum_{k=1}^{m} b_k \chi_{B_k},$$

then

$$\sum_{i=1}^{n} a_i \mu(A_i) = \sum_{k=1}^{m} b_k \mu(B_k).$$

Proof. Consider the common refinement of the partitions $\{A_i\}$ and $\{B_k\}$. Let

$$C_{ik} = A_i \cap B_k.$$

On the set C_{ik}, φ takes the value a_i and also b_k. Thus, if $C_{ik} \neq \varnothing$, we must have

$$a_i = b_k.$$

Therefore

$$\begin{aligned}
\sum_i a_i \mu(A_i) &= \sum_i a_i \sum_k \mu(A_i \cap B_k) \\
&= \sum_{i,k} a_i \mu(C_{ik}) \\
&= \sum_{i,k} b_k \mu(C_{ik}) \\
&= \sum_k b_k \mu(B_k).
\end{aligned}$$

□

Proposition 9.6 (Linearity for Simple Functions)**.** *Let φ, ψ be simple measurable functions and $\alpha, \beta \in \mathbb{R}$. Then:*

$$\int_E (\alpha\varphi + \beta\psi)\, d\mu = \alpha \int_E \varphi\, d\mu + \beta \int_E \psi\, d\mu.$$

Additional Proof. Take a common measurable refinement $\{C_{ij}\}$ on which both φ and ψ are constant. On C_{ij},

$$\varphi = a_i, \qquad \psi = b_j,$$

so

$$\alpha\varphi + \beta\psi = \alpha a_i + \beta b_j.$$

Therefore

$$\begin{aligned}
\int_E (\alpha\varphi + \beta\psi)\, d\mu &= \sum_{i,j} (\alpha a_i + \beta b_j) \mu(E \cap C_{ij}) \\
&= \alpha \sum_{i,j} a_i \mu(E \cap C_{ij}) + \beta \sum_{i,j} b_j \mu(E \cap C_{ij}) \\
&= \alpha \int_E \varphi\, d\mu + \beta \int_E \psi\, d\mu.
\end{aligned}$$

□

9.2 The Integral of Non-negative Functions

We extend the integral to general non-negative measurable functions by approximating them from below with simple functions.

Definition 9.7. Let

$$f : \Omega \to [0, \infty]$$

be a measurable function. We define the Lebesgue integral of f over E as:

$$\int_E f \, d\mu = \sup \left\{ \int_E \varphi \, d\mu : 0 \leq \varphi \leq f, \ \varphi \text{ is simple} \right\}.$$

Lemma 9.8 (Monotone Approximation)**.** *Let $f \geq 0$ be measurable. There exists a sequence of simple functions $\{\varphi_n\}$ such that:*

1. $$0 \leq \varphi_1 \leq \varphi_2 \leq \cdots \leq f.$$
2. $$\varphi_n(x) \to f(x)$$

   pointwise for all x.
3. *If f is bounded, convergence is uniform.*

Construction. Standard "dyadic" construction:

$$\varphi_n(x) = \begin{cases} \dfrac{k-1}{2^n}, & \dfrac{k-1}{2^n} \leq f(x) < \dfrac{k}{2^n}, \quad k = 1, \ldots, n2^n, \\ n, & f(x) \geq n. \end{cases}$$

Remark 9.9. The construction approximates f from below while simultaneously controlling both the vertical step size and the truncation height. The mesh size is 2^{-n} and the truncation level is n.

Additional Details. For every x,

$$0 \leq \varphi_n(x) \leq f(x).$$

As n increases, the dyadic partition becomes finer and the truncation level increases, so

$$\varphi_n(x) \leq \varphi_{n+1}(x).$$

If $f(x) < \infty$, then for all sufficiently large n we have $f(x) < n$, and the construction gives

$$0 \leq f(x) - \varphi_n(x) < 2^{-n}.$$

Hence

$$\varphi_n(x) \to f(x).$$

If $f(x) = \infty$, then

$$\varphi_n(x) = n$$

for all sufficiently large n, so again

$$\varphi_n(x) \to \infty = f(x).$$

If f is bounded by M, then for $n > M$,

$$0 \leq f(x) - \varphi_n(x) < 2^{-n}$$

for every x, and therefore

$$\sup_x |f(x) - \varphi_n(x)| \leq 2^{-n} \to 0.$$

Thus the convergence is uniform. □

Proposition 9.10 (Properties for Non-negative Functions)**.** *Let $f, g \geq 0$ be measurable.*

1. ***Monotonicity:*** *If $f \leq g$ a.e., then*
$$\int f \leq \int g.$$
2. ***Homogeneity:*** *For $c \geq 0$,*
$$\int cf = c \int f.$$
3. ***Additivity:***
$$\int (f + g) = \int f + \int g.$$
(Requires Monotone Convergence Theorem to prove rigorously).
4. ***Chebyshev's Inequality:*** *For $c > 0$,*
$$\mu(\{x : f(x) \geq c\}) \leq \frac{1}{c} \int f \, d\mu.$$

Proof of Chebyshev. Let

$$E_c = \{x : f(x) \geq c\}.$$

Define simple function

$$\varphi = c \cdot \chi_{E_c}.$$

Clearly

$$0 \leq \varphi \leq f$$

on Ω. By definition of the integral:

$$\int_\Omega \varphi \, d\mu \leq \int_\Omega f \, d\mu \Longrightarrow c \cdot \mu(E_c) \leq \int f \, d\mu.$$

Corollary 9.11. *If $f \geq 0$ and*

$$\int f \, d\mu = 0,$$

then

$$f = 0$$

almost everywhere.

Proof. For every $c > 0$, Chebyshev's inequality gives

$$\mu(\{x : f(x) \geq c\}) \leq \frac{1}{c} \int f \, d\mu = 0.$$

In particular, for every $n \in \mathbb{N}$,

$$\mu\left(\left\{x : f(x) \geq \frac{1}{n}\right\}\right) = 0.$$

Since

$$\{x : f(x) > 0\} = \bigcup_{n=1}^{\infty} \left\{x : f(x) \geq \frac{1}{n}\right\},$$

countable subadditivity gives

$$\mu(\{x : f(x) > 0\}) = 0.$$

Hence $f = 0$ almost everywhere. □

9.3 The General Lebesgue Integral

For a general measurable function

$$f : \Omega \to \mathbb{R}$$

(which may change sign), we split it into positive and negative parts.

Definition 9.12. Define the positive part f^+ and negative part f^-:

$$f^+(x) = \max\{f(x), 0\}, \qquad f^-(x) = \max\{-f(x), 0\}.$$

Note that

$$f = f^+ - f^-, \qquad |f| = f^+ + f^-.$$

Both f^+ and f^- are non-negative measurable functions.

Definition 9.13 (Integrability)**.** A measurable function f is said to be Lebesgue integrable (or summable) on Ω if both

$$\int f^+ \qquad \text{and} \qquad \int f^-$$

are finite. In this case, we define:

$$\int_\Omega f \, d\mu = \int_\Omega f^+ \, d\mu - \int_\Omega f^- \, d\mu.$$

We denote the space of integrable functions by

$$L^1(\Omega, \mu).$$

If one of the integrals is $+\infty$ and the other is finite, the integral is defined as $+\infty$ or $-\infty$. If both are $+\infty$, the integral is undefined.

Proposition 9.14. *f is integrable if and only if $|f|$ is integrable. That is:*

$$f \in L^1(\mu) \iff \int_\Omega |f| \, d\mu < \infty.$$

Furthermore,

$$\left|\int_\Omega f \, d\mu\right| \leq \int_\Omega |f| \, d\mu.$$

Additional Proof. Since

$$|f| = f^+ + f^-,$$

we have

$$\int |f| = \int f^+ + \int f^-.$$

Thus

$$\int |f| < \infty$$

if and only if both

$$\int f^+ < \infty \qquad \text{and} \qquad \int f^- < \infty.$$

This is exactly the definition of integrability of f.

Moreover,

$$\left|\int f\right| = \left|\int f^+ - \int f^-\right| \leq \int f^+ + \int f^- = \int |f|.$$

□

9.4 Basic Properties

Let

$$f, g \in L^1(\Omega, \mu)$$

and $\alpha, \beta \in \mathbb{R}$.

Theorem 9.15 (Linearity)**.** *The space $L^1(\Omega, \mu)$ is a vector space, and the integral is a linear functional:*

$$\int_E (\alpha f + \beta g)\, d\mu = \alpha \int_E f\, d\mu + \beta \int_E g\, d\mu.$$

Sketch for Additivity. If

$$h = f + g,$$

then

$$h^+ - h^- = f^+ - f^- + g^+ - g^-.$$

Rearranging terms to have only positive sums:

$$h^+ + f^- + g^- = h^- + f^+ + g^+.$$

Integrating both sides (using additivity for non-negative functions):

$$\int h^+ + \int f^- + \int g^- = \int h^- + \int f^+ + \int g^+.$$

Rearranging terms back yields

$$\int h = \int f + \int g.$$

Remark 9.16. The rearrangement above is legitimate because all the terms

$$\int h^+, \quad \int h^-, \quad \int f^+, \quad \int f^-, \quad \int g^+, \quad \int g^-$$

are finite for integrable functions.

Theorem 9.17 (Additivity over Domains)**.** *If A and B are disjoint measurable sets, then*

$$\int_{A\cup B} f\,d\mu = \int_A f\,d\mu + \int_B f\,d\mu.$$

More generally, if

$$E = \bigsqcup_{n=1}^{\infty} E_n$$

is a countable disjoint union, and f is integrable on E, then

$$\int_E f\,d\mu = \sum_{n=1}^{\infty} \int_{E_n} f\,d\mu.$$

This implies that the set function

$$\nu(A) = \int_A f\,d\mu$$

is a measure (specifically, a signed measure).

Remark 9.18. The construction

$$\nu(A) = \int_A f\,d\mu$$

illustrates three points:

1. For a non-negative function $f \geq 0$, ν is an ordinary non-negative measure.
2. For a general integrable function f, ν may take both positive and negative values and is therefore a signed measure.
3. The decomposition

   $$f = f^+ - f^-$$

   induces the corresponding decomposition

$$\nu(A) = \int_A f^+\,d\mu - \int_A f^-\,d\mu.$$

Theorem 9.19 (Absolute Continuity of the Integral)**.** *Let f be integrable on Ω. For every $\varepsilon > 0$, there exists $\delta > 0$ such that for any measurable set $E \subset \Omega$:*

$$\mu(E) < \delta \Longrightarrow \left|\int_E f\,d\mu\right| < \varepsilon.$$

Proof. **Step 1: Prove for bounded functions.** If

$$|f| \leq M,$$

then

$$\left|\int_E f\right| \leq \int_E |f| \leq M\mu(E).$$

Take

$$\delta = \frac{\varepsilon}{M}.$$

Step 2: General case. Since

$$f \in L^1,$$

$|f|$ is integrable. Let

$$f_n = \min(|f|, n).$$

By Monotone Convergence,

$$\int f_n \to \int |f|.$$

Choose N large enough so that

$$\int (|f| - f_N) < \frac{\varepsilon}{2}.$$

Let

$$\delta = \frac{\varepsilon}{2N}.$$

If

$$\mu(E) < \delta,$$

then

$$\begin{aligned} \int_E |f| &= \int_E (|f| - f_N) + \int_E f_N \\ &< \frac{\varepsilon}{2} + N\mu(E) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Therefore

$$\left| \int_E f \, d\mu \right| \leq \int_E |f| \, d\mu < \varepsilon.$$

□

Chapter 10

Convergence Theorems

10 Convergence Theorems

One of the primary deficiencies of the Riemann integral is its poor behavior under limits. The Lebesgue integral resolves this, provided we have control over the sequence (either monotonicity or domination).

10.1 Lebesgue Monotone Convergence Theorem (Levi)

Theorem 10.1 (Levi's Theorem / Monotone Convergence Theorem)**.** *Let $(\Omega, \mathcal{A}, \mu)$ be a measure space. Let $\{f_n\}_{n=1}^{\infty}$ be a sequence of measurable functions such that:*

1. $$0 \leq f_1(x) \leq f_2(x) \leq \cdots$$
   for almost every $x \in \Omega$ (Monotonicity).
2. $$f_n(x) \to f(x)$$
   almost everywhere.

Then f is measurable (with values in $[0, \infty]$) and:

$$\lim_{n\to\infty} \int_\Omega f_n \, d\mu = \int_\Omega f \, d\mu.$$

Proof. Since

$$f_n \leq f_{n+1},$$

the limit $f(x)$ exists (possibly $+\infty$) and is measurable. By monotonicity of the integral,

$$\int f_n \leq \int f_{n+1} \leq \int f.$$

Thus,

$$L := \lim_{n\to\infty} \int_\Omega f_n \, d\mu \leq \int_\Omega f \, d\mu.$$

To prove the reverse inequality

$$\int f \leq L,$$

let φ be any simple function such that

$$0 \leq \varphi \leq f.$$

Let

$$c \in (0, 1).$$

Consider the sets

$$E_n = \{x \in \Omega : f_n(x) \geq c\varphi(x)\}.$$

Since

$$f_n \uparrow f$$

and $c < 1$, the sequence of sets E_n is increasing:

$$E_n \subset E_{n+1}.$$

Moreover,

$$\bigcup_{n=1}^{\infty} E_n = \Omega$$

on the support of φ.

Therefore

$$\int_\Omega f_n \, d\mu \geq \int_{E_n} f_n \, d\mu \geq c \int_{E_n} \varphi \, d\mu.$$

Taking the limit as $n \to \infty$,

$$L \geq c \lim_{n\to\infty} \int_{E_n} \varphi \, d\mu = c \int_\Omega \varphi \, d\mu.$$

Since this holds for every $c < 1$, letting $c \uparrow 1$ gives

$$L \geq \int_\Omega \varphi \, d\mu.$$

Taking the supremum over all simple functions

$$0 \leq \varphi \leq f,$$

we obtain

$$L \geq \int_\Omega f \, d\mu.$$

Hence

$$\lim_{n\to\infty} \int_\Omega f_n \, d\mu = \int_\Omega f \, d\mu.$$

□

Remark 10.2. The essential point is that for a non-negative increasing sequence,

$$f_n \uparrow f,$$

the limit and the integral can be interchanged without any dominating function:

$$\int \lim_{n\to\infty} f_n = \lim_{n\to\infty} \int f_n.$$

Corollary 10.3 (Series of Non-negative Functions)**.** *If $g_k \geq 0$ are measurable, then*

$$\int_\Omega \sum_{k=1}^{\infty} g_k \, d\mu = \sum_{k=1}^{\infty} \int_\Omega g_k \, d\mu.$$

(Integration and summation can be swapped for non-negative terms).

Additional Proof. Let

$$f_n = \sum_{k=1}^{n} g_k.$$

Then

$$0 \leq f_1 \leq f_2 \leq \cdots$$

and

$$f_n(x) \to \sum_{k=1}^{\infty} g_k(x).$$

By the Monotone Convergence Theorem,

$$\int_\Omega \sum_{k=1}^{\infty} g_k \, d\mu = \lim_{n\to\infty} \int_\Omega \sum_{k=1}^{n} g_k \, d\mu.$$

By finite linearity,

$$\int_\Omega \sum_{k=1}^{n} g_k \, d\mu = \sum_{k=1}^{n} \int_\Omega g_k \, d\mu.$$

Hence

$$\int_\Omega \sum_{k=1}^{\infty} g_k \, d\mu = \sum_{k=1}^{\infty} \int_\Omega g_k \, d\mu.$$

□

10.2 Fatou's Lemma

When the sequence is not monotone, we cannot guarantee convergence of integrals, but we have an inequality for the lim inf.

Theorem 10.4 (Fatou's Lemma)**.** *Let $\{f_n\}$ be a sequence of non-negative measurable functions. Then:*

$$\int_\Omega \liminf_{n\to\infty} f_n \, d\mu \le \liminf_{n\to\infty} \int_\Omega f_n \, d\mu.$$

Proof. Let

$$g_k(x) = \inf_{n\ge k} f_n(x).$$

Then g_k is measurable,

$$g_k \ge 0,$$

and

$$g_k(x) \le f_n(x)$$

for all $n \ge k$.

This implies

$$\int g_k \le \int f_n$$

for all $n \ge k$, so

$$\int g_k \le \inf_{n\ge k} \int f_n.$$

The sequence $\{g_k\}$ is non-decreasing:

$$g_k \le g_{k+1},$$

and

$$g_k(x) \to \liminf_{n\to\infty} f_n(x).$$

By the Monotone Convergence Theorem:

$$\begin{aligned}
\int \liminf_{n\to\infty} f_n &= \int \lim_{k\to\infty} g_k \\
&= \lim_{k\to\infty} \int g_k \\
&\le \lim_{k\to\infty} \left(\inf_{n\ge k} \int f_n \right) \\
&= \liminf_{n\to\infty} \int f_n.
\end{aligned}$$

□

Remark 10.5. Fatou's Lemma is one-sided. In general,

$$\int \liminf f_n$$

can be strictly smaller than

$$\liminf \int f_n.$$

Thus some "mass" may disappear in the pointwise limit.

Example 10.6 (Strict Inequality)**.** Let

$$f_n = n\chi_{[0,1/n]}$$

on $\mathbb{R}$.

- Pointwise limit:

  $$f_n(x) \to 0$$

  for all $x > 0$ (and at 0 if handled), so

  $$\liminf f_n = 0.$$

  Hence

  $$\int 0 = 0.$$

- Integrals:

  $$\int f_n = n \cdot \frac{1}{n} = 1.$$

  Thus

  $$\liminf \int f_n = 1.$$

- Therefore

  $$0 < 1,$$

  so the inequality is strict. (Mass "escapes" to infinity or concentrates at a point).

10.3 Lebesgue Dominated Convergence Theorem

This is the most powerful tool for swapping limits and integrals.

Theorem 10.7 (Dominated Convergence Theorem (DCT))**.** *Let $\{f_n\}$ be a sequence of measurable functions on Ω such that*

$$f_n(x) \to f(x)$$

almost everywhere. Suppose there exists an integrable function

$$g \in L^1(\Omega)$$

(the dominating function) such that:

$$|f_n(x)| \leq g(x) \qquad \text{a.e. for all } n.$$

Then f is integrable and

$$\lim_{n\to\infty} \int_\Omega f_n \, d\mu = \int_\Omega f \, d\mu.$$

(Also implies convergence in L^1 norm:

$$\int_\Omega |f_n - f| \, d\mu \to 0.$$

)

Standard Proof using Fatou. Since

$$|f_n| \leq g$$

and

$$f_n \to f \quad \text{a.e.},$$

we have

$$|f| \leq g \quad \text{a.e.},$$

so

$$f \in L^1.$$

Consider the non-negative functions:

$$h_n = g + f_n \geq 0$$

and

$$k_n = g - f_n \geq 0.$$

Applying Fatou's Lemma to h_n:

$$\int (g + f) \leq \liminf_{n\to\infty} \int (g + f_n).$$

Thus

$$\int g + \int f \leq \int g + \liminf_{n\to\infty} \int f_n,$$

and therefore

$$\int f \leq \liminf_{n\to\infty} \int f_n.$$

Applying Fatou's Lemma to k_n:

$$\int (g - f) \leq \liminf_{n\to\infty} \int (g - f_n).$$

Hence

$$\int g - \int f \leq \int g - \limsup_{n\to\infty} \int f_n,$$

so

$$\limsup_{n\to\infty} \int f_n \leq \int f.$$

Therefore

$$\limsup_{n\to\infty} \int f_n \leq \int f \leq \liminf_{n\to\infty} \int f_n.$$

Since always

$$\liminf a_n \leq \limsup a_n,$$

the only possibility is

$$\lim_{n\to\infty} \int f_n = \int f.$$

□

Additional Proof of L^1 Convergence. Since

$$f_n \to f \quad \text{a.e.},$$

we have

$$|f_n - f| \to 0 \quad \text{a.e.}$$

Moreover,

$$|f_n - f| \leq |f_n| + |f| \leq 2g.$$

Since

$$2g \in L^1,$$

applying the Dominated Convergence Theorem to $|f_n - f|$ gives

$$\int_\Omega |f_n - f|\, d\mu \to 0.$$

Thus

$$\|f_n - f\|_{L^1} \to 0.$$

□

Remark 10.8. The dominating function g is the key hypothesis. Pointwise or almost everywhere convergence alone is not enough to exchange limit and integral. DCT provides an integrable envelope:

$$|f_n| \leq g \in L^1,$$

which prevents mass from escaping or concentrating uncontrollably.

Remark 10.9 (Proof in Notes via Egorov). The lecture notes also sketch a proof using Egorov's Theorem (assuming $\mu(\Omega) < \infty$ or using absolute continuity of integral). Key idea: Split integral into set E where convergence is uniform (by Egorov) and E^c where measure is small.

$$\left| \int f_n - \int f \right| \leq \int_E |f_n - f| + \int_{E^c} |f_n - f|.$$

The first term vanishes by uniform convergence. The second term is bounded by

$$\int_{E^c} 2g,$$

which is small by absolute continuity of the integral of g.

10.4 Relationship with Riemann Integral

Theorem 10.10 (Consistency)**.** *Let*

$$f : [a, b] \to \mathbb{R}$$

be bounded and Riemann integrable. Then f is Lebesgue integrable and the values coincide:

$$^R\int_a^b f(x)\, dx = {}^L\int_{[a,b]} f\, d\mu.$$

Remark 10.11. If the notation

$$\overset{R}{\int} \qquad \text{and} \qquad \overset{L}{\int}$$

causes a compilation error in your setup, replace the displayed formula by

$$\int_a^b f(x)\,dx = \int_{[a,b]} f\,d\mu,$$

with the sentence stating that the left-hand side is the Riemann integral and the right-hand side is the Lebesgue integral.

Theorem 10.12 (Lebesgue's Criterion for Riemann Integrability)**.** *Let*

$$f : [a, b] \to \mathbb{R}$$

be a bounded function. f is Riemann integrable if and only if f is continuous almost everywhere (with respect to Lebesgue measure). i.e.,

$$\mu\left(\{x \in [a, b] : f \text{ is discontinuous at } x\}\right) = 0.$$

Outline. Define the oscillation of f at x:

$$\omega_f(x) = \lim_{\delta \to 0} \sup\left\{|f(y) - f(z)| : y, z \in (x - \delta, x + \delta) \cap [a, b]\right\}.$$

f is continuous at x if and only if

$$\omega_f(x) = 0.$$

Let

$$D_\varepsilon = \{x : \omega_f(x) \geq \varepsilon\}.$$

The set of discontinuities is

$$D = \bigcup_{k=1}^{\infty} D_{1/k}.$$

Using the definition of Riemann sums (Darboux sums $U(P, f)$ and $L(P, f)$), one can show that the condition

$$U(P, f) - L(P, f) < \varepsilon$$

for sufficiently fine partitions is equivalent to the set D having measure zero.

Example 10.13.

- **Dirichlet Function:**

  $$\chi_{\mathbb{Q}}$$

  on $[0, 1]$. Discontinuous everywhere. Not Riemann integrable. Lebesgue integral is 0.
- **Thomae's Function (Popcorn function):**

  $$f(p/q) = \frac{1}{q}, \qquad f(\text{irrational}) = 0.$$

  Continuous at irrationals (a.e.). Thus Riemann integrable (integral is 0).

Chapter 11

Infinite and Improper Integrals

11 Infinite and Improper Integrals

11.1 Integration over Sets of Infinite Measure

Definition 11.1 (σ-finite Space)**.** A measure space $(\Omega, \mathcal{A}, \mu)$ is σ-finite if

$$\Omega = \bigcup_{j=1}^{\infty} X_j$$

with

$$\mu(X_j) < \infty.$$

Remark 11.2. The condition of σ-finiteness means that although the whole space may have infinite measure, it can be exhausted by countably many measurable pieces of finite measure. For example,

$$\mathbb{R} = \bigcup_{n=1}^{\infty} [-n, n],$$

and every interval $[-n, n]$ has finite Lebesgue measure. Thus $\mathbb{R}$ with Lebesgue measure is σ-finite.

Definition 11.3. For $f \geq 0$ on a σ-finite space, we define

$$\int_\Omega f \, d\mu = \sup_n \int_{X_n} f \, d\mu$$

or

$$\sum_{j=1}^{\infty} \int_{X_j \setminus X_{j-1}} f \, d\mu.$$

The value is independent of the decomposition. For general f, integrability requires

$$\int_\Omega |f| \, d\mu < \infty.$$

Remark 11.4. When using

$$\int_\Omega f \, d\mu = \sup_n \int_{X_n} f \, d\mu,$$

we may take the sets X_n to form an increasing exhaustion

$$X_1 \subset X_2 \subset \cdots, \qquad \bigcup_{n=1}^{\infty} X_n = \Omega.$$

If the original σ-finite decomposition is not increasing, replace it by

$$Y_n = \bigcup_{j=1}^{n} X_j.$$

Then

$$Y_n \uparrow \Omega.$$

Additional Explanation of Independence. Let

$$X_n \uparrow \Omega.$$

For $f \geq 0$, define

$$f_n = f\chi_{X_n}.$$

Then

$$0 \leq f_1 \leq f_2 \leq \cdots$$

and

$$f_n(x) \to f(x)$$

for every $x \in \Omega$.

By the Monotone Convergence Theorem,

$$\int_\Omega f\,d\mu = \lim_{n\to\infty} \int_\Omega f_n\,d\mu = \lim_{n\to\infty} \int_{X_n} f\,d\mu.$$

Since the sequence of integrals is increasing,

$$\lim_{n\to\infty} \int_{X_n} f\,d\mu = \sup_n \int_{X_n} f\,d\mu.$$

Therefore the resulting value depends only on f and Ω, not on the particular increasing exhaustion.

If we define

$$E_1 = X_1, \qquad E_j = X_j \setminus X_{j-1}, \quad j \geq 2,$$

then

$$\Omega = \bigsqcup_{j=1}^{\infty} E_j.$$

Since $f \geq 0$, countable additivity of the integral gives

$$\int_\Omega f\,d\mu = \sum_{j=1}^{\infty} \int_{E_j} f\,d\mu = \sum_{j=1}^{\infty} \int_{X_j \setminus X_{j-1}} f\,d\mu.$$

□

11.2 Improper Riemann vs. Lebesgue Integral

The classical Improper Riemann Integral is defined via limits:

$$\int_0^\infty f(x)\,dx := \lim_{R\to\infty} \int_0^R f(x)\,dx.$$

Proposition 11.5. *1. If $f \geq 0$, then the Improper Riemann integral exists (finite)*

$$\Longleftrightarrow$$

f is Lebesgue integrable on $[0, \infty)$.

2. If f changes sign, Lebesgue integrability requires absolute convergence.

Additional Explanation. For $f \geq 0$, define

$$f_R(x) = f(x)\chi_{[0,R]}(x).$$

As $R \to \infty$,

$$f_R(x) \uparrow f(x).$$

Therefore, by the Monotone Convergence Theorem,

$$\int_0^\infty f\,d\mu = \lim_{R\to\infty}\int_0^R f(x)\,dx.$$

Thus the non-negative improper Riemann integral and the Lebesgue integral agree whenever either side is finite.

For a function which changes sign, write

$$f = f^+ - f^-.$$

Lebesgue integrability means

$$\int f^+ < \infty \qquad \text{and} \qquad \int f^- < \infty.$$

Equivalently,

$$\int |f| = \int f^+ + \int f^- < \infty.$$

Thus Lebesgue integrability requires absolute convergence rather than merely conditional convergence. □

Example 11.6 (Sinc Function)**.** Consider

$$f(x) = \frac{\sin x}{x}$$

on $[0,\infty)$.

- Riemann:
$$\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$$
(converges conditionally).
- Lebesgue:
$$\int_0^\infty \left|\frac{\sin x}{x}\right| dx = \infty.$$

Proof of Divergence:

$$\int_{k\pi}^{(k+1)\pi} \frac{|\sin x|}{x}\,dx \geq \frac{1}{(k+1)\pi}\int_0^\pi \sin t\,dt = \frac{2}{(k+1)\pi}.$$

Since

$$\sum_{k=1}^\infty \frac{1}{k}$$

diverges, the integral diverges.

Conclusion:

$$f \notin L^1([0,\infty)).$$

The Lebesgue integral is undefined (or $+\infty - \infty$). This highlights that the Lebesgue integral is an absolute integral.

Remark 11.7. The point of this example is the distinction

$$\boxed{\text{conditional convergence}} \qquad \text{vs.} \qquad \boxed{\text{Lebesgue integrability}}.$$

The improper Riemann integral may exist because positive and negative oscillations cancel:

$$\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}.$$

But the Lebesgue theory defines

$$\int f = \int f^+ - \int f^-$$

only when this subtraction is meaningful. For $\sin x/x$, both positive and negative masses are infinite, so one encounters the indeterminate expression

$$+\infty - \infty.$$

Remark 11.8. For functions on bounded intervals with singularities (e.g., $x^{-1/2}$ on $(0,1]$), if $f \geq 0$, improper Riemann and Lebesgue integrals coincide.

Example 11.9 (Additional Explanation)**.** For

$$f(x) = x^{-1/2}, \qquad x \in (0,1],$$

the improper Riemann integral is

$$\int_0^1 x^{-1/2}\,dx = \lim_{\varepsilon \downarrow 0} \int_\varepsilon^1 x^{-1/2}\,dx = \lim_{\varepsilon \downarrow 0} 2(1-\sqrt{\varepsilon}) = 2.$$

Since $f \geq 0$, the Lebesgue integral gives the same value:

$$\int_{(0,1]} x^{-1/2}\,d\mu = 2.$$

Thus an unbounded function can still be Lebesgue integrable.

Chapter 12

Signed Measures (Charges)

12 Signed Measures (Charges)

Standard measures are non-negative. However, in applications (e.g., integrating a function that changes sign, electric charge distribution), it is useful to allow measures to take negative values.

12.1 Definition and Hahn Decomposition

Definition 12.1 (Signed Measure / Charge)**.** Let $(\Omega, \mathcal{A})$ be a measurable space. A set function

$$\nu : \mathcal{A} \to \mathbb{R} \cup \{+\infty, -\infty\}$$

is called a signed measure (or a charge) if:

1. $$\nu(\varnothing) = 0.$$
2. ν assumes at most one of the values $+\infty$ or $-\infty$ (to avoid $\infty - \infty$).
3. σ-additivity: For any countable collection of disjoint sets

$$\{E_j\}_{j=1}^{\infty} \subset \mathcal{A},$$

$$\nu\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} \nu(E_j),$$

where the series converges absolutely if $\nu(\bigcup E_j)$ is finite.

Remark 12.2. The essential difference between an ordinary measure and a signed measure is that

$$\mu(E) \geq 0$$

for an ordinary measure, while $\nu(E)$ may be positive or negative. The σ-additivity property is retained.

Example 12.3. Let μ be a measure and $f \in L^1(\mu)$ (or semi-integrable). Define

$$\nu(E) = \int_E f \, d\mu.$$

Then ν is a signed measure.

Additional Proof. Clearly,

$$\nu(\varnothing) = \int_{\varnothing} f \, d\mu = 0.$$

Let $\{E_j\}_{j=1}^{\infty}$ be pairwise disjoint. Since

$$\chi_{\bigcup_{j=1}^{\infty} E_j} = \sum_{j=1}^{\infty} \chi_{E_j},$$

we have

$$f\chi_{\bigcup_{j=1}^{\infty} E_j} = \sum_{j=1}^{\infty} f\chi_{E_j}.$$

Hence, by countable additivity of the integral,

$$\begin{aligned}\nu\left(\bigcup_{j=1}^{\infty} E_j\right) &= \int_{\bigcup_{j=1}^{\infty} E_j} f\, d\mu \\ &= \sum_{j=1}^{\infty} \int_{E_j} f\, d\mu \\ &= \sum_{j=1}^{\infty} \nu(E_j).\end{aligned}$$

Thus ν is σ-additive. □

Definition 12.4 (Positive and Negative Sets)**.** Let ν be a signed measure.

- A set $P \in \mathcal{A}$ is called positive with respect to ν if for every measurable $E \subset P$,

$$\nu(E) \geq 0.$$

- A set $N \in \mathcal{A}$ is called negative with respect to ν if for every measurable $E \subset N$,

$$\nu(E) \leq 0.$$

- A set K is a null set if it is both positive and negative (i.e., every subset has measure 0).

Remark 12.5. Notice that the condition for a positive set is stronger than merely

$$\nu(P) \geq 0.$$

We require

$$\nu(E) \geq 0$$

for *every* measurable subset $E \subset P$. Similarly,

$$\nu(N) \leq 0$$

alone does not imply that N is a negative set.

Theorem 12.6 (Hahn Decomposition Theorem)**.** *Let ν be a signed measure on $(\Omega, \mathcal{A})$. There exist disjoint sets P and N such that:*

1. $$\Omega = P \cup N.$$
2. *P is a positive set for ν.*
3. *N is a negative set for ν.*

This decomposition is essentially unique: if P', N' is another such pair, then

$$P \triangle P' \qquad \text{and} \qquad N \triangle N'$$

are null sets.

Proof Sketch. (Based on lecture notes strategy)

1. Eliminate the infinite case (assume ν does not take $-\infty$).

2. Define
$$\lambda = \sup\{\nu(A) : A \text{ is a negative set}\}?$$
No, standard proof usually defines
$$\alpha = \inf \nu(A).$$

3. The notes use an exhaustion argument. First, prove that if
$$\nu(A) < 0,$$
there exists a negative subset
$$B \subset A$$
with
$$\nu(B) \leq \nu(A).$$

4. Construct N as the union of a sequence of negative sets extracting the "negative mass" of ν. Then
$$P = \Omega \setminus N.$$

Additional Detailed Proof for the Finite Case. Assume first that ν is finite. Let
$$\alpha = \inf\{\nu(A) : A \in \mathcal{A}\}.$$
Choose measurable sets E_n such that
$$\nu(E_n) < \alpha + 2^{-n}.$$
The difficulty is that the E_n need not themselves be negative sets. We therefore use the standard extraction lemma: if a measurable set A is not negative, then there exists a measurable subset $B \subset A$ with
$$\nu(B) > 0.$$
Repeatedly removing such positive subsets produces a negative subset while controlling the measure removed.

Applying this construction to sets whose measures approach α, one obtains negative sets N_n such that
$$\nu(N_n) \to \alpha.$$
A countable union of negative sets is negative: indeed, after replacing them by the disjoint sets
$$N_1, \quad N_2 \setminus N_1, \quad N_3 \setminus (N_1 \cup N_2), \ldots,$$
every measurable subset has non-positive ν-measure by σ-additivity.

Thus
$$N = \bigcup_{n=1}^{\infty} N_n$$
is negative. Set
$$P = \Omega \setminus N.$$
If P were not positive, there would exist a measurable
$$E \subset P$$

with

$$\nu(E) < 0.$$

Extracting a negative subset from E and adjoining it to N would produce a negative set with strictly smaller measure than the infimum α, a contradiction. Therefore P is positive.

Hence

$$\Omega = P \cup N, \qquad P \cap N = \varnothing,$$

with P positive and N negative. □

Remark 12.7. The Hahn decomposition splits the *space* according to the sign of the signed measure:

$$\boxed{\Omega = P \sqcup N}.$$

On P, every measurable subset has non-negative ν-measure; on N, every measurable subset has non-positive ν-measure.

12.2 Jordan Decomposition

The Hahn decomposition allows us to split the space. The Jordan decomposition allows us to split the measure itself.

Definition 12.8 (Mutually Singular Measures)**.** Two measures (or signed measures) μ and ν are mutually singular (denoted

$$\mu \perp \nu$$

) if there exist disjoint sets A, B such that

$$\Omega = A \cup B$$

and

$$\mu(B) = 0 \qquad \text{and} \qquad \nu(A) = 0.$$

(i.e., μ lives on A, ν lives on B).

Remark 12.9. The phrase "μ lives on A" means that the complement B carries no μ-mass:

$$\mu(B) = 0.$$

Similarly, "ν lives on B" means

$$\nu(A) = 0.$$

Thus mutually singular measures are concentrated on disjoint measurable parts of the space.

Theorem 12.10 (Jordan Decomposition)**.** *Let ν be a signed measure with Hahn decomposition*

$$\Omega = P \cup N.$$

Define two non-negative measures:

$$\nu^+(E) := \nu(E \cap P) \qquad \textit{and} \qquad \nu^-(E) := -\nu(E \cap N).$$

Then:

1.

$$\nu = \nu^+ - \nu^-.$$

2.

$$\nu^+ \perp \nu^-.$$

3. If

$$\nu = \mu_1 - \mu_2$$

is any other decomposition into non-negative measures, then

$$\nu^+ \leq \mu_1 \qquad \textit{and} \qquad \nu^- \leq \mu_2$$

(Jordan decomposition is minimal).

Additional Details. Since

$$E = (E \cap P) \sqcup (E \cap N),$$

we have

$$\nu(E) = \nu(E \cap P) + \nu(E \cap N).$$

By definition,

$$\nu(E \cap P) = \nu^+(E)$$

and

$$\nu(E \cap N) = -\nu^-(E).$$

Hence

$$\nu(E) = \nu^+(E) - \nu^-(E).$$

Furthermore,

$$\nu^+(N) = \nu(N \cap P) = 0,$$

and

$$\nu^-(P) = -\nu(P \cap N) = 0.$$

Thus

$$\nu^+ \perp \nu^-.$$

Now suppose

$$\nu = \mu_1 - \mu_2$$

with μ_1, μ_2 non-negative. If $E \subset P$, then

$$\nu^+(E) = \nu(E) = \mu_1(E) - \mu_2(E) \leq \mu_1(E).$$

For general E,

$$\nu^+(E) = \nu^+(E \cap P) \leq \mu_1(E \cap P) \leq \mu_1(E).$$

Hence

$$\nu^+ \leq \mu_1.$$

Similarly,

$$\nu^- \leq \mu_2.$$

□

Remark 12.11. For the important example

$$\nu(E) = \int_E f \, d\mu,$$

the Jordan decomposition corresponds exactly to

$$f = f^+ - f^-.$$

Indeed,

$$\nu^+(E) = \int_E f^+ \, d\mu, \qquad \nu^-(E) = \int_E f^- \, d\mu.$$

Thus the Jordan decomposition of a signed measure is the measure-theoretic analogue of decomposing a real-valued function into its positive and negative parts.

Definition 12.12 (Total Variation)**.** The total variation of ν is the measure $|\nu|$ defined by:

$$|\nu|(E) = \nu^+(E) + \nu^-(E).$$

Equivalently,

$$|\nu|(E) = \sup \sum_j |\nu(E_j)|$$

over all partitions of E. ν is called finite (or bounded) if

$$|\nu|(\Omega) < \infty.$$

Remark 12.13. For

$$\nu(E) = \int_E f \, d\mu,$$

we have

$$|\nu|(E) = \int_E |f| \, d\mu.$$

This follows from

$$|f| = f^+ + f^-.$$

Thus total variation plays for signed measures the same role that the absolute value plays for functions.

12.3 Absolute Continuity of Measures

Definition 12.14. Let ν be a signed measure and μ be a positive measure on $(\Omega, \mathcal{A})$. We say ν is absolutely continuous with respect to μ (denoted

$$\nu \ll \mu$$

) if for every $E \in \mathcal{A}$:

$$\mu(E) = 0 \Longrightarrow \nu(E) = 0.$$

Remark 12.15. The notation

$$\nu \ll \mu$$

means that ν cannot assign mass to a set that is invisible to μ. Equivalently, every μ-null set must also be a ν-null set.

For example, if

$$\nu(E) = \int_E f \, d\mu,$$

then automatically

$$\nu \ll \mu,$$

because

$$\mu(E) = 0 \Longrightarrow \int_E f \, d\mu = 0.$$

Proposition 12.16 ($\varepsilon - \delta$ Characterization)**.** *If ν is finite, then*

$$\nu \ll \mu$$

if and only if:

$$\forall \varepsilon > 0, \ \exists \delta > 0, \ \textit{s.t.} \ \mu(E) < \delta \Longrightarrow |\nu(E)| < \varepsilon.$$

Additional Proof. Suppose first that the ε–δ condition holds. If

$$\mu(E) = 0,$$

then

$$\mu(E) < \delta$$

for every $\delta > 0$. Therefore

$$|\nu(E)| < \varepsilon$$

for every $\varepsilon > 0$, which implies

$$\nu(E) = 0.$$

Hence

$$\nu \ll \mu.$$

Conversely, suppose

$$\nu \ll \mu$$

and assume the ε–δ statement is false. Then there exists $\varepsilon_0 > 0$ and a sequence of measurable sets E_n such that

$$\mu(E_n) < 2^{-n}$$

but

$$|\nu(E_n)| \geq \varepsilon_0.$$

Let

$$F_n = \bigcup_{k=n}^{\infty} E_k.$$

Then

$$\mu(F_n) \leq \sum_{k=n}^{\infty} 2^{-k} \to 0.$$

The sequence F_n is decreasing, so

$$F = \bigcap_{n=1}^{\infty} F_n$$

satisfies

$$\mu(F) = 0.$$

Since

$$\nu \ll \mu,$$

we obtain

$$\nu(F) = 0.$$

Using finiteness of ν and continuity of a finite signed measure along decreasing sets,

$$\nu(F_n) \to \nu(F) = 0.$$

This contradicts the existence of sets inside the tails carrying uniformly non-negligible ν-mass. Hence the ε–δ condition must hold. □

Chapter 13

The Radon-Nikodym Theorem

13 The Radon-Nikodym Theorem

This theorem is the fundamental theorem of calculus for measure theory. It states that absolute continuity implies representation as an integral.

13.1 The Theorem and Proof

Theorem 13.1 (Radon-Nikodym Theorem)**.** *Let $(\Omega, \mathcal{A}, \mu)$ be a σ-finite measure space, and let ν be a σ-finite signed measure (or charge). If*

$$\nu \ll \mu,$$

then there exists a measurable function f (unique μ-a.e.) such that for all $E \in \mathcal{A}$:

$$\nu(E) = \int_E f \, d\mu.$$

The function f is called the Radon-Nikodym derivative and denoted

$$f = \frac{d\nu}{d\mu}.$$

Remark 13.2. The notation

$$\frac{d\nu}{d\mu}$$

should be viewed as a measure-theoretic analogue of an ordinary derivative. The identity

$$\nu(E) = \int_E \frac{d\nu}{d\mu} \, d\mu$$

says that the measure ν has a "density" with respect to μ.

Proof for finite measures (Supremum Approach). Assume

$$\mu(\Omega) < \infty$$

and ν is a non-negative finite measure (general case follows by Jordan decomposition).

Step 1: Construct the candidate function.

Let $\mathcal{S}$ be the set of all non-negative measurable functions g such that:

$$\int_E g \, d\mu \leq \nu(E), \qquad \forall E \in \mathcal{A}.$$

$\mathcal{S}$ is non-empty (contains 0). Let

$$\alpha = \sup \left\{ \int_\Omega g \, d\mu : g \in \mathcal{S} \right\}.$$

Since

$$\nu(\Omega) < \infty,$$

α is finite.

Choose a sequence

$$g_n \in \mathcal{S}$$

such that

$$\int_\Omega g_n \, d\mu \to \alpha.$$

Let

$$f_n = \max(g_1, \ldots, g_n).$$

One can verify

$$f_n \in \mathcal{S}$$

(proof: split Ω into sets where each g_i is maximal). The sequence f_n is increasing. Let

$$f = \lim_{n\to\infty} f_n.$$

By Monotone Convergence Theorem:

$$\int_E f\, d\mu = \lim_{n\to\infty} \int_E f_n\, d\mu \le \nu(E).$$

Thus

$$f \in \mathcal{S}$$

and attains the supremum:

$$\int_\Omega f\, d\mu = \alpha.$$

Additional Detail: why $f_n \in \mathcal{S}$. It is enough to prove that if

$$g, h \in \mathcal{S},$$

then

$$\max(g, h) \in \mathcal{S}.$$

Let

$$A = \{x : g(x) \ge h(x)\}.$$

Then

$$\max(g, h) = g\chi_A + h\chi_{A^c}.$$

For any measurable E,

$$\begin{aligned}\int_E \max(g, h)\, d\mu &= \int_{E\cap A} g\, d\mu + \int_{E\cap A^c} h\, d\mu \\ &\le \nu(E \cap A) + \nu(E \cap A^c) \\ &= \nu(E).\end{aligned}$$

Hence

$$\max(g, h) \in \mathcal{S}.$$

Applying this repeatedly gives

$$f_n = \max(g_1, \ldots, g_n) \in \mathcal{S}.$$

□

Step 2: Analysis of the remainder.

Define a measure λ_s by

$$\lambda_s(E) = \nu(E) - \int_E f\, d\mu.$$

Since

$$f \in \mathcal{S},$$

we have

$$\lambda_s \geq 0.$$

We claim

$$\lambda_s = 0.$$

Suppose

$$\lambda_s(\Omega) > 0.$$

Since

$$\mu(\Omega) < \infty,$$

μ cannot be singular to λ_s in a trivial way? No, we use

$$\nu \ll \mu.$$

Claim: λ_s is singular to μ? No. Actually, we assume

$$\nu \ll \mu.$$

We want to show

$$\lambda_s = 0.$$

Suppose not. Then there exists $\varepsilon > 0$ and a set P with

$$\mu(P) > 0$$

such that

$$\lambda_s - \varepsilon\mu$$

is a positive measure on P (using Hahn decomposition on the signed measure $\lambda_s - \varepsilon\mu$). i.e., for

$$E \subset P,$$

$$\lambda_s(E) \geq \varepsilon\mu(E).$$

Then:

$$\begin{aligned}\nu(E) &= \int_E f\,d\mu + \lambda_s(E) \\ &\geq \int_E f\,d\mu + \varepsilon\mu(E) \\ &= \int_E (f+\varepsilon)\,d\mu.\end{aligned}$$

This implies

$$h = f + \varepsilon\chi_P$$

is in $\mathcal{S}$. But

$$\int_\Omega h\,d\mu = \int_\Omega f\,d\mu + \varepsilon\mu(P) > \alpha,$$

contradiction!

Therefore,

$$\lambda_s = 0,$$

so

$$\nu(E) = \int_E f\,d\mu.$$

Remark 13.3. The proof is a maximality argument. We construct the largest density f satisfying

$$\int_E f \, d\mu \leq \nu(E)$$

for all measurable E. If a positive remainder were left over, then on some positive-μ set we could increase f slightly, contradicting maximality.

Proposition 13.4 (Uniqueness of the Radon-Nikodym Derivative)**.** *If f and g both satisfy*

$$\nu(E) = \int_E f \, d\mu = \int_E g \, d\mu$$

for every $E \in \mathcal{A}$, then

$$f = g \qquad \mu\text{-a.e.}$$

Proof. For every measurable set E,

$$\int_E (f - g) \, d\mu = 0.$$

Let

$$A = \{x : f(x) > g(x)\}.$$

Then

$$A = \bigcup_{n=1}^{\infty} \left\{ x : f(x) - g(x) \geq \frac{1}{n} \right\}.$$

If

$$\mu(A) > 0,$$

then for some n,

$$\mu\left(\left\{ f - g \geq \frac{1}{n} \right\} \right) > 0.$$

Taking this set as E gives

$$\int_E (f - g) \, d\mu \geq \frac{1}{n} \mu(E) > 0,$$

a contradiction. Hence

$$\mu(\{f > g\}) = 0.$$

Similarly,

$$\mu(\{g > f\}) = 0.$$

Therefore

$$f = g \qquad \mu\text{-a.e.}$$

□

13.2 Lebesgue Decomposition

What if ν is not absolutely continuous with respect to μ?

Theorem 13.5 (Lebesgue Decomposition)**.** *Let ν and μ be σ-finite measures. There exists a unique pair of measures*

$$(\nu_{ac}, \nu_s)$$

such that:

1.

$$\nu = \nu_{ac} + \nu_s.$$

2.

$$\nu_{ac} \ll \mu$$

(Absolutely continuous part).

3.

$$\nu_s \perp \mu$$

(Singular part).

Furthermore, ν_{ac} has the form

$$\int f \, d\mu$$

by Radon-Nikodym.

Remark 13.6. Thus every suitable measure ν can be decomposed into two conceptually different parts:

$$\boxed{\nu = \nu_{ac} + \nu_s}.$$

The absolutely continuous part has a density:

$$\nu_{ac}(E) = \int_E \frac{d\nu_{ac}}{d\mu} \, d\mu,$$

while the singular part is concentrated on a set of μ-measure zero.

Connection to R-N proof. In the proof above, if we do not assume

$$\nu \ll \mu,$$

the remainder measure λ_s constructed (the maximal remainder) turns out to be exactly the singular part ν_s. The argument showing

$$\lambda_s = 0$$

relied on

$$\nu \ll \mu.$$

Without that, we can only prove

$$\lambda_s \perp \mu.$$

Example 13.7. Let μ be Lebesgue measure on $\mathbb{R}$, and δ_0 be the Dirac mass at 0. Let

$$\nu = \mu + \delta_0.$$

Then:

- $$\nu_{ac} = \mu$$

  (density $f = 1$).

- $$\nu_s = \delta_0$$

  (supported on $\{0\}$, while

  $$\mu(\{0\}) = 0).$$

Remark 13.8. This example makes the decomposition concrete:

$$\nu(E) = \mu(E) + \delta_0(E).$$

The first term is spread continuously with respect to Lebesgue measure, whereas the second term places a unit point mass at 0.

There is no ordinary function f satisfying

$$\delta_0(E) = \int_E f \, d\mu,$$

because

$$\delta_0 \not\ll \mu.$$

Indeed,

$$\mu(\{0\}) = 0 \qquad \text{but} \qquad \delta_0(\{0\}) = 1.$$

Chapter 14

Differentiation and Integration

14 Differentiation and Integration

This chapter addresses the inverse relationship between differentiation and integration. In classical calculus, the Fundamental Theorem of Calculus holds for C^1 functions. In the Lebesgue theory, we seek the necessary and sufficient conditions on a function f such that

$$f(x) = f(a) + \int_a^x f'(t)\,dt.$$

This leads to the concepts of Bounded Variation and Absolute Continuity.

14.1 Monotone Functions and Differentiation

Theorem 14.1 (Lebesgue's Differentiation Theorem for Monotone Functions)**.** *Let*

$$f : [a, b] \to \mathbb{R}$$

be a monotone increasing function. Then:

1. *f is differentiable almost everywhere on $[a, b]$.*
2. *The derivative f' is measurable and non-negative.*
3. *f' is Lebesgue integrable, and satisfies the inequality:*

$$\int_a^b f'(x)\,dx \leq f(b) - f(a).$$

Remark 14.2. The inequality can be strict. For the Cantor-Lebesgue function K, we have

$$K'(x) = 0 \quad \text{a.e.},$$

so

$$\int_0^1 K'(x)\,dx = 0,$$

but

$$K(1) - K(0) = 1.$$

The "missing" part of the integral corresponds to the "singular" nature of the function (it rises on a set of measure zero).

Proof Idea for the Inequality. Extend f to $\mathbb{R}$ by setting

$$f(x) = f(b)$$

for $x > b$. Define the difference quotient:

$$D_n(x) = n\left[f\left(x + \frac{1}{n}\right) - f(x)\right].$$

Since f is monotone,

$$D_n \geq 0.$$

We know

$$D_n(x) \to f'(x)$$

a.e. By Fatou's Lemma:

$$\int_a^b f'(x)\,dx = \int_a^b \liminf_{n\to\infty} D_n(x)\,dx \leq \liminf_{n\to\infty} \int_a^b D_n(x)\,dx.$$

Calculating the integral of D_n:

$$\int_a^b n\left[f\left(x+\frac{1}{n}\right)-f(x)\right]dx = n\int_b^{b+1/n} f(x)\,dx - n\int_a^{a+1/n} f(x)\,dx.$$

As $n \to \infty$, this converges to

$$f(b) - f(a)$$

(using continuity of integral or simple bounds).

Remark 14.3. The theorem says that monotonicity is already strong enough to guarantee differentiability almost everywhere, even though a monotone function need not be continuously differentiable or even continuous everywhere.

14.2 Functions of Bounded Variation (BV)

Definition 14.4 (Total Variation)**.** Let

$$f : [a, b] \to \mathbb{R}.$$

A partition P of $[a, b]$ is a set of points

$$a = x_0 < x_1 < \cdots < x_n = b.$$

The variation of f with respect to P is:

$$V(f, P) = \sum_{j=1}^{n} |f(x_j) - f(x_{j-1})|.$$

The Total Variation of f on $[a, b]$ is:

$$V_a^b(f) = \sup\{V(f, P) : P \text{ is a partition of } [a, b]\}.$$

If

$$V_a^b(f) < \infty,$$

we say f is of Bounded Variation (BV).

Remark 14.5. Geometrically, total variation measures the total amount that the graph moves vertically, counting both upward and downward motion. Thus it is generally much larger than

$$|f(b) - f(a)|.$$

For a monotone function, however, there is no cancellation between upward and downward oscillations, so the two quantities coincide.

Example 14.6. 1. If f is monotone on $[a, b]$, then

$$V_a^b(f) = |f(b) - f(a)|,$$

so

$$f \in BV.$$

2. If f is Lipschitz continuous (e.g., C^1 with bounded derivative), then
$$f \in BV.$$
3. Consider
$$f(x) = x \sin(1/x)$$
for $x \neq 0$ and
$$f(0) = 0.$$
f is continuous but not BV on $[0, 1]$ (the oscillations sum to ∞).
4.
$$f(x) = x^2 \sin(1/x)$$
is BV (the derivative is bounded).

Additional Detail for the Lipschitz Case. Suppose
$$|f(x) - f(y)| \leq L|x - y|$$
for all $x, y \in [a, b]$. For any partition
$$P = \{x_0, \ldots, x_n\},$$
we have
$$\begin{aligned} V(f, P) &= \sum_{j=1}^{n} |f(x_j) - f(x_{j-1})| \\ &\leq L \sum_{j=1}^{n} (x_j - x_{j-1}) \\ &= L(b - a). \end{aligned}$$
Taking the supremum over all partitions gives
$$V_a^b(f) \leq L(b - a) < \infty.$$
Hence
$$f \in BV[a, b].$$
□

Proposition 14.7 (Properties). *1. If*
$$f \in BV,$$
then f is bounded.
2. *The space $BV[a, b]$ is a vector space (closed under sum and scalar multiplication) and an algebra (closed under product).*
3. ***Additivity:** If*
$$a < c < b,$$
then
$$V_a^b(f) = V_a^c(f) + V_c^b(f).$$

Additional Details. For (1), fix $x \in [a,b]$. Then

$$|f(x) - f(a)| \le V_a^b(f).$$

Hence

$$|f(x)| \le |f(a)| + V_a^b(f),$$

so f is bounded.

For sums,

$$\begin{aligned} V(f+g,P) &= \sum_j |(f+g)(x_j) - (f+g)(x_{j-1})| \\ &\le V(f,P) + V(g,P). \end{aligned}$$

Thus

$$V_a^b(f+g) \le V_a^b(f) + V_a^b(g).$$

Similarly,

$$V_a^b(cf) = |c| V_a^b(f).$$

For products,

$$fg(x_j) - fg(x_{j-1}) = f(x_j)\Big(g(x_j) - g(x_{j-1})\Big) + g(x_{j-1})\Big(f(x_j) - f(x_{j-1})\Big).$$

Since BV functions are bounded, summing gives a finite bound for

$$V_a^b(fg).$$

For additivity, any partition of $[a,b]$ may be refined by inserting c, and then the corresponding variation splits into the variation on $[a,c]$ and $[c,b]$. Taking suprema yields

$$V_a^b(f) = V_a^c(f) + V_c^b(f).$$

□

Theorem 14.8 (Jordan Decomposition)**.** *A function f is of bounded variation on $[a,b]$ if and only if it can be written as the difference of two monotone increasing functions:*

$$f(x) = g(x) - h(x),$$

where g, h are increasing.

Proof. If

$$f = g - h,$$

both are BV, so f is BV.

Conversely, define the variation function

$$v(x) = V_a^x(f).$$

It is easy to show that $v(x)$ is increasing. Let

$$g(x) = v(x)$$

and

$$h(x) = v(x) - f(x).$$

Check h is increasing: Let $x < y$.

$$\begin{aligned} h(y) - h(x) &= (V_a^y(f) - V_a^x(f)) - (f(y) - f(x)) \\ &= V_x^y(f) - (f(y) - f(x)). \end{aligned}$$

Since

$$|f(y) - f(x)| \leq V_x^y(f),$$

we have

$$h(y) - h(x) \geq 0.$$

□

Corollary 14.9. *If*

$$f \in BV[a, b],$$

then f is differentiable almost everywhere.

Additional Proof. By the Jordan decomposition,

$$f = g - h,$$

where g and h are monotone increasing. By Lebesgue's Differentiation Theorem for monotone functions, both g and h are differentiable almost everywhere. Therefore

$$f' = g' - h'$$

exists almost everywhere. □

14.3 Absolute Continuity (AC)

We seek a condition stronger than continuity (and BV) that eliminates "singular" behavior like the Cantor function.

Definition 14.10 (Absolute Continuity)**.** A function

$$f : [a, b] \to \mathbb{R}$$

is absolutely continuous (AC) on $[a, b]$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for any finite collection of disjoint sub-intervals

$$(a_k, b_k) \subset [a, b] :$$

$$\sum_{k=1}^{n} (b_k - a_k) < \delta \Longrightarrow \sum_{k=1}^{n} |f(b_k) - f(a_k)| < \varepsilon.$$

Remark 14.11. Ordinary uniform continuity controls the image of *one* short interval. Absolute continuity requires simultaneous control over an arbitrary finite collection of disjoint intervals whose *total length* is small.

Proposition 14.12. *1.*

$$AC \Longrightarrow \textit{Uniformly Continuous}$$

(take $n = 1$).

2.

$$AC \Longrightarrow BV.$$

3. If f is Lipschitz, then f is AC.

Additional Proof. For (1), given $\varepsilon > 0$, let $\delta > 0$ be supplied by absolute continuity. Taking a single interval (x, y) gives

$$|x - y| < \delta \Longrightarrow |f(x) - f(y)| < \varepsilon.$$

Hence f is uniformly continuous.

For (2), choose $\delta > 0$ corresponding to $\varepsilon = 1$. Let

$$a = x_0 < x_1 < \cdots < x_n = b$$

be any partition. Divide the interval $[a, b]$ into finitely many groups of consecutive subintervals so that the total length of the intervals in each group is less than δ. The number of groups can be bounded independently of the partition by a constant depending only on $b - a$ and δ. On each group, absolute continuity implies that the total variation is less than 1. Hence

$$V(f, P)$$

is bounded uniformly over all partitions P, and therefore

$$f \in BV[a, b].$$

For (3), suppose

$$|f(x) - f(y)| \leq L|x - y|.$$

Then for disjoint intervals (a_k, b_k),

$$\sum_{k=1}^{n} |f(b_k) - f(a_k)| \leq L \sum_{k=1}^{n} (b_k - a_k).$$

Choosing

$$\delta = \frac{\varepsilon}{L}$$

proves absolute continuity. □

Example 14.13 (Cantor Function is not AC)**.** The Cantor function K is continuous and BV. However, consider the intervals removed in the construction of the Cantor set. Let $\{(a_k, b_k)\}$ be the 2^n intervals removed at step n. The sum of lengths is $\to 1$. Wait, let's look at the remaining intervals (inside the Cantor set construction) or simply use the Fundamental Theorem.

Direct check: K maps the Cantor set C (measure 0) to the full range $[0, 1]$ (measure 1). A key property of AC functions is the Lusin N-property:

$$\mu(E) = 0 \Longrightarrow \mu(f(E)) = 0.$$

K fails this.

Theorem 14.14. *Let f be AC on $[a, b]$. Then f is of bounded variation, and f maps sets of measure zero to sets of measure zero.*

Additional Proof of the Null-Set Property. Let

$$E \subset [a, b]$$

satisfy

$$\mu(E) = 0.$$

Fix $\varepsilon > 0$. By absolute continuity, choose $\delta > 0$ such that any finite disjoint family of intervals with total length less than δ satisfies

$$\sum_k |f(b_k) - f(a_k)| < \varepsilon.$$

Since E has measure zero, it can be covered by a countable collection of open intervals $\{I_k\}$ whose total length is less than δ. After decomposing the union into disjoint open intervals, absolute continuity gives that the total length of the corresponding image intervals is less than ε. Thus the outer measure of $f(E)$ is at most ε.

Since $\varepsilon > 0$ is arbitrary,

$$\mu(f(E)) = 0.$$

□

14.4 The Fundamental Theorem of Lebesgue Calculus

This theorem characterizes AC functions as exactly those that are indefinite integrals.

Theorem 14.15 (Part 1: Differentiation of the Integral)**.** *Let*

$$g \in L^1([a, b]).$$

Define

$$f(x) = \int_a^x g(t)\, dt.$$

Then f is absolutely continuous on $[a, b]$, and

$$f'(x) = g(x)$$

almost everywhere.

Idea. Absolute continuity follows directly from the absolute continuity of the measure

$$\nu(E) = \int_E g.$$

The differentiability

$$f' = g$$

relies on the Lebesgue Differentiation Theorem (differentiation of averages).

Additional Detail for Absolute Continuity. Let

$$\varepsilon > 0.$$

Since

$$g \in L^1([a, b]),$$

the absolute continuity of the integral gives a $\delta > 0$ such that

$$\mu(E) < \delta \Longrightarrow \int_E |g| < \varepsilon.$$

For any finite collection of disjoint intervals (a_k, b_k) satisfying

$$\sum_{k=1}^{n} (b_k - a_k) < \delta,$$

we have

$$\begin{aligned} \sum_{k=1}^{n} |f(b_k) - f(a_k)| &= \sum_{k=1}^{n} \left| \int_{a_k}^{b_k} g(t)\, dt \right| \\ &\leq \sum_{k=1}^{n} \int_{a_k}^{b_k} |g(t)|\, dt \\ &= \int_{\bigcup_k (a_k, b_k)} |g(t)|\, dt \\ &< \varepsilon. \end{aligned}$$

Hence f is absolutely continuous. □

Theorem 14.16 (Part 2: The Newton-Leibniz Formula)**.** *A function*

$$f : [a, b] \to \mathbb{R}$$

is absolutely continuous if and only if:

1. *f is differentiable almost everywhere.*
2. $$f' \in L^1([a, b]).$$
3. *For all*
$$x \in [a, b],$$
$$f(x) - f(a) = \int_a^x f'(t)\, dt.$$

Additional Explanation of the Converse. Suppose f is absolutely continuous. Since

$$AC \Longrightarrow BV,$$

f is differentiable almost everywhere.

Define

$$F(x) = \int_a^x f'(t)\, dt.$$

By Part 1, F is absolutely continuous and

$$F' = f' \qquad \text{a.e.}$$

Hence

$$H = f - F$$

is absolutely continuous and

$$H' = 0 \qquad \text{a.e.}$$

The key property of absolutely continuous functions is that if an AC function has derivative zero almost everywhere, then it is constant. Therefore

$$H(x) = H(a) = f(a).$$

Thus

$$f(x) - f(a) = F(x) = \int_a^x f'(t)\,dt.$$

□

Remark 14.17. This is exactly where absolute continuity differs from mere continuity or bounded variation. The Cantor function satisfies

$$K' = 0 \qquad \text{a.e.}$$

but is not constant. This cannot happen for an absolutely continuous function.

Observation 14.18.

- **Monotone:** f' exists a.e.,

$$\int f' \le f(b) - f(a).$$

- **BV:** f' exists a.e. (difference of monotone).
- **AC:** f' exists a.e. and

$$\int f' = f(b) - f(a).$$

  This is the "good" class for calculus.
- **Singular:** f is continuous, non-constant, but

$$f' = 0$$

  a.e. (e.g., Cantor).

$$f \in BV \iff f = f_{AC} + f_{\text{sing}}.$$

Chapter 15

The L^p Spaces

15 The L^p Spaces

We introduce the Lebesgue spaces L^p. These are Banach spaces of functions defined by integral norms. They play a central role in functional analysis and PDEs.

15.1 Definitions and Inequalities

Definition 15.1 (L^p Norm)**.** Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and

$$1 \le p < \infty.$$

The space $L^p(\Omega)$ consists of all measurable functions

$$f : \Omega \to \mathbb{R}$$

(modulo equality a.e.) such that:

$$\|f\|_p := \left(\int_\Omega |f|^p \, d\mu \right)^{1/p} < \infty.$$

For $p = \infty$, $L^\infty(\Omega)$ is the space of essentially bounded functions:

$$\|f\|_\infty := \operatorname{ess\,sup} |f| = \inf\{C \ge 0 : |f(x)| \le C \text{ a.e.}\}.$$

Remark 15.2. Functions in L^p are identified modulo equality almost everywhere. Thus if

$$f = g \qquad \text{a.e.},$$

then they represent the same element of L^p and

$$\|f - g\|_p = 0.$$

This is necessary because otherwise $\| \cdot \|_p$ would only be a seminorm on measurable functions.

Lemma 15.3 (Young's Inequality)**.** *Let $a, b \ge 0$ and*

$$1 < p, q < \infty$$

with

$$\frac{1}{p} + \frac{1}{q} = 1$$

(conjugate exponents). Then:

$$ab \le \frac{a^p}{p} + \frac{b^q}{q}.$$

Equality holds if and only if

$$a^p = b^q.$$

Additional Proof. Set

$$\lambda = \frac{1}{p}, \qquad 1 - \lambda = \frac{1}{q}.$$

Since $\log x$ is concave,

$$\log(\lambda x_1 + (1-\lambda)x_2) \ge \lambda \log x_1 + (1-\lambda) \log x_2.$$

Exponentiating,

$$\lambda x_1 + (1-\lambda)x_2 \geq x_1^{\lambda} x_2^{1-\lambda}.$$

Take

$$x_1 = a^p, \qquad x_2 = b^q.$$

Then

$$x_1^{1/p} x_2^{1/q} = ab,$$

so

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}.$$

Equality in the concavity inequality occurs exactly when

$$x_1 = x_2,$$

hence

$$a^p = b^q.$$

□

Theorem 15.4 (Hölder's Inequality)**.** *Let*

$$f \in L^p$$

and

$$g \in L^q$$

with

$$\frac{1}{p} + \frac{1}{q} = 1.$$

Then the product

$$fg$$

is in L^1 *and:*

$$\|fg\|_1 = \int_\Omega |fg| \, d\mu \leq \|f\|_p \|g\|_q.$$

Proof. If

$$\|f\|_p = 0$$

or

$$\|g\|_q = 0,$$

the result is trivial. Assume they are non-zero. Normalize the functions: let

$$F = \frac{|f|}{\|f\|_p} \qquad \text{and} \qquad G = \frac{|g|}{\|g\|_q}.$$

Apply Young's inequality pointwise:

$$F(x)G(x) \leq \frac{F(x)^p}{p} + \frac{G(x)^q}{q}.$$

Integrate both sides:

$$\int FG \, d\mu \leq \frac{1}{p} \int F^p \, d\mu + \frac{1}{q} \int G^q \, d\mu = \frac{1}{p} \cdot 1 + \frac{1}{q} \cdot 1 = 1.$$

Substitute back definitions of F and G to get the result. □

Remark 15.5. The conjugacy relation

$$\frac{1}{p}+\frac{1}{q}=1$$

is exactly what makes the normalization in Young's inequality compatible with the L^p and L^q norms.

Theorem 15.6 (Minkowski's Inequality)**.** *For*

$$1 \le p \le \infty$$

and

$$f, g \in L^p :$$
$$\|f+g\|_p \le \|f\|_p + \|g\|_p.$$

This establishes that $\|\cdot\|_p$ satisfies the triangle inequality and is indeed a norm.

Case $1 < p < \infty$.

$$|f+g|^p \le (|f|+|g|)|f+g|^{p-1} = |f||f+g|^{p-1} + |g||f+g|^{p-1}.$$

Integrate and apply Hölder with exponents p and q

$$\left(q = \frac{p}{p-1}\right) :$$

$$\int |f+g|^p \le \|f\|_p \left\||f+g|^{p-1}\right\|_q + \|g\|_p \left\||f+g|^{p-1}\right\|_q .$$

Note that

$$\begin{aligned}\left\||f+g|^{p-1}\right\|_q &= \left(\int |f+g|^{(p-1)q}\right)^{1/q} \\ &= \left(\int |f+g|^p\right)^{1/q} \\ &= \|f+g\|_p^{p/q}.\end{aligned}$$

Dividing both sides by

$$\|f+g\|_p^{p/q}$$

(assuming non-zero) yields the result, since

$$p - \frac{p}{q} = 1.$$

Additional Details for $p = 1$ *and* $p = \infty$. For $p = 1$,

$$|f+g| \le |f| + |g|.$$

Integrating,

$$\|f+g\|_1 \le \|f\|_1 + \|g\|_1.$$

For $p = \infty$, we have almost everywhere

$$|f+g| \le |f| + |g| \le \|f\|_\infty + \|g\|_\infty.$$

Hence

$$\|f+g\|_\infty \le \|f\|_\infty + \|g\|_\infty.$$

□

Remark 15.7. Minkowski's inequality is the L^p analogue of the ordinary triangle inequality:

$$\boxed{\|f+g\|_p \le \|f\|_p + \|g\|_p}.$$

Together with positivity and homogeneity, this is what makes L^p a normed space.

15.2 Completeness (Riesz-Fischer Theorem)

A normed space is a Banach space if it is complete (every Cauchy sequence converges).

Theorem 15.8 (Riesz-Fischer)**.** *For*

$$1 \leq p \leq \infty,$$

the space

$$L^p(\Omega)$$

is complete.

Proof for $p < \infty$. Let $\{f_n\}$ be a Cauchy sequence in L^p.
Step 1: Extract a fast-converging subsequence. For each $k \in \mathbb{N}$, choose n_k such that for all

$$m, n \geq n_k,$$
$$\|f_n - f_m\|_p < 2^{-k}.$$

We can ensure

$$n_{k+1} > n_k.$$

Thus

$$\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}.$$

Step 2: Construct a pointwise limit. Consider the series

$$g(x) = \sum_{k=1}^{\infty} |f_{n_{k+1}}(x) - f_{n_k}(x)|.$$

By the monotone convergence theorem (applied to partial sums) and Minkowski inequality:

$$\|g\|_p \leq \sum_{k=1}^{\infty} \|f_{n_{k+1}} - f_{n_k}\|_p < \sum_{k=1}^{\infty} 2^{-k} = 1.$$

Thus

$$g \in L^p,$$

which implies

$$g(x) < \infty$$

almost everywhere.

Consequently, the series

$$\sum_{k=1}^{\infty} (f_{n_{k+1}} - f_{n_k})$$

converges absolutely a.e. This is a telescoping series, so the sequence

$$\{f_{n_k}(x)\}$$

converges pointwise a.e. to some function $f(x)$.
Step 3: Convergence in L^p. We have

$$f_{n_k} \to f$$

a.e. Also,

$$|f - f_{n_k}| = \left| \sum_{j=k}^{\infty} (f_{n_{j+1}} - f_{n_j}) \right| \leq \sum_{j=k}^{\infty} |f_{n_{j+1}} - f_{n_j}| \leq g.$$

Since

$$g \in L^p,$$

we can apply the Dominated Convergence Theorem to

$$|f_{n_k} - f|^p.$$

Indeed,

$$|f_{n_k} - f|^p \to 0$$

pointwise a.e. and

$$|f_{n_k} - f|^p \leq g^p,$$

where

$$g^p \in L^1.$$

Thus

$$\|f_{n_k} - f\|_p^p = \int |f_{n_k} - f|^p \, d\mu \to 0,$$

so

$$\|f_{n_k} - f\|_p \to 0.$$

Since the original sequence is Cauchy and has a convergent subsequence, the whole sequence converges to f. □

Remark 15.9. The central trick in the Riesz-Fischer proof is not to work directly with the original Cauchy sequence, but to extract a subsequence satisfying

$$\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}.$$

The summability of

$$\sum 2^{-k}$$

then forces almost-everywhere absolute convergence of the telescoping series.

15.3 Separability and Approximation

Theorem 15.10 (Approximation)**.** *Let*

$$1 \leq p < \infty.$$

1. *The set of simple functions (integrable) is dense in*

$$L^p(\Omega).$$

2. *If*

$$\Omega = \mathbb{R}^n$$

(or nice domain), the set of continuous functions with compact support

$$C_c(\mathbb{R}^n)$$

is dense in

$$L^p(\mathbb{R}^n).$$

Additional Explanation for Simple Functions. Let

$$f \in L^p.$$

Since

$$|f|^p \in L^1,$$

we can first truncate:

$$f_N = \max(-N, \min(f, N)).$$

Then

$$|f - f_N|^p \to 0$$

pointwise, and

$$|f - f_N|^p \leq |f|^p.$$

By the Dominated Convergence Theorem,

$$\|f - f_N\|_p \to 0.$$

For each bounded measurable f_N, use the standard dyadic construction to approximate it pointwise by simple functions φ_m with

$$|\varphi_m - f_N| \leq 2^{-m}.$$

On a finite-measure support this immediately implies L^p convergence; in general one also truncates the support to sets of finite measure. Combining the two approximations gives simple functions converging to f in L^p. □

Corollary 15.11. *$L^p(\mathbb{R}^n)$ is separable for*

$$1 \leq p < \infty.$$

(Consider simple functions with rational coefficients on dyadic intervals). L^∞ is generally not separable.

Remark 15.12. The restriction

$$p < \infty$$

is essential for the separability statement. In L^∞, one can often construct an uncountable family of functions separated by a fixed positive distance in the essential-supremum norm, preventing the existence of a countable dense subset.

15.4 The Dual of L^p

Theorem 15.13 (Riesz Representation for L^p)**.** *Let*

$$1 < p < \infty$$

and q be the conjugate exponent. There is an isometric isomorphism between

$$(L^p)^*$$

(the continuous linear functionals on L^p) and

$$L^q.$$

Every bounded linear functional

$$\ell : L^p \to \mathbb{R}$$

is of the form:

$$\ell(f) = \int_\Omega fg \, d\mu,$$

for a unique

$$g \in L^q.$$

And

$$\|\ell\|_{(L^p)^*} = \|g\|_q.$$

Additional Verification of One Direction. Let

$$g \in L^q$$

and define

$$\ell_g(f) = \int_\Omega fg \, d\mu.$$

By Hölder's inequality,

$$|\ell_g(f)| \leq \|f\|_p \|g\|_q.$$

Hence ℓ_g is a bounded linear functional and

$$\|\ell_g\| \leq \|g\|_q.$$

To obtain the reverse inequality, for $g \neq 0$ define

$$f = \frac{\operatorname{sgn}(g)|g|^{q-1}}{\|g\|_q^{q/p}}.$$

Since

$$(q-1)p = q,$$

we have

$$\|f\|_p = 1.$$

Then

$$\ell_g(f) = \frac{\int |g|^q \, d\mu}{\|g\|_q^{q/p}} = \|g\|_q.$$

Therefore

$$\|\ell_g\| = \|g\|_q.$$

The converse statement—that every bounded functional arises this way—is the substantive Riesz representation theorem. □

Remark 15.14. For

$$p = 1,$$

$$(L^1)^* = L^\infty$$

under the usual hypotheses (for example, σ-finite measure spaces). However,

$$(L^\infty)^* \neq L^1$$

in general (the dual of L^∞ contains measures that are not absolutely continuous).

Remark 15.15. The basic duality pattern is

$$\boxed{(L^p)^* \simeq L^q, \qquad \frac{1}{p} + \frac{1}{q} = 1, \qquad 1 < p < \infty.}$$

The pairing is

$$\langle f, g \rangle = \int f g \, d\mu,$$

and Hölder's inequality is precisely what guarantees that this pairing is bounded.

Chapter 16

Product Measures

16 Product Measures

We aim to define integration on product spaces $\Omega_1 \times \Omega_2$, reducing double integrals to iterated integrals.

16.1 Product σ-algebras

Let $(\Omega_1, \mathcal{A}_1, \mu_1)$ and $(\Omega_2, \mathcal{A}_2, \mu_2)$ be measure spaces. The cartesian product

$$\Omega = \Omega_1 \times \Omega_2.$$

The family of measurable rectangles is:

$$\mathcal{R} = \{A \times B : A \in \mathcal{A}_1,\ B \in \mathcal{A}_2\}.$$

$\mathcal{R}$ is a semi-ring. The product σ-algebra

$$\mathcal{A}_1 \otimes \mathcal{A}_2$$

is the σ-algebra generated by $\mathcal{R}$.

Remark 16.1. The notation

$$\mathcal{A}_1 \otimes \mathcal{A}_2$$

does not mean an algebraic tensor product here. It denotes the smallest σ-algebra on $\Omega_1 \times \Omega_2$ containing every measurable rectangle

$$A \times B.$$

16.2 Construction of Product Measure

Definition 16.2 (Product Measure)**.** Define a set function on rectangles:

$$\pi(A \times B) = \mu_1(A) \cdot \mu_2(B).$$

This extends to a measure on the ring generated by rectangles, and by Carathéodory's Theorem, extends to a measure

$$\mu = \mu_1 \times \mu_2$$

on

$$\mathcal{A}_1 \otimes \mathcal{A}_2.$$

If μ_1, μ_2 are σ-finite, the product measure μ is unique.

Remark 16.3. On a measurable rectangle, product measure behaves exactly like ordinary area or volume:

$$(\mu_1 \times \mu_2)(A \times B) = \mu_1(A)\mu_2(B).$$

For Lebesgue measure on $\mathbb{R}^2$, this reduces to

$$\text{area} = \text{length} \times \text{width}.$$

16.3 Fubini's Theorem and Tonelli's Theorem

Let

$$(\Omega, \mathcal{A}, \mu) = (\Omega_1 \times \Omega_2, \mathcal{A}_1 \otimes \mathcal{A}_2, \mu_1 \times \mu_2)$$

be the product of two σ-finite measure spaces. Let

$$f(x, y)$$

be a measurable function on Ω.

Theorem 16.4 (Tonelli's Theorem (Non-negative Functions))**.** *If*

$$f : \Omega \to [0, \infty]$$

is measurable, then the iterated integrals are measurable and equal to the product integral:

$$\int_\Omega f \, d\mu = \int_{\Omega_1} \left(\int_{\Omega_2} f(x, y) \, d\mu_2(y) \right) d\mu_1(x)$$

and

$$\int_\Omega f \, d\mu = \int_{\Omega_2} \left(\int_{\Omega_1} f(x, y) \, d\mu_1(x) \right) d\mu_2(y).$$

Therefore,

$$\int_{\Omega_1} \left(\int_{\Omega_2} f(x, y) \, d\mu_2(y) \right) d\mu_1(x) = \int_{\Omega_2} \left(\int_{\Omega_1} f(x, y) \, d\mu_1(x) \right) d\mu_2(y).$$

(The value may be $+\infty$).

Remark 16.5. For Tonelli's theorem, the crucial assumption is

$$f \geq 0.$$

No integrability assumption is required. Thus for a non-negative measurable function, the order of integration can be exchanged even if the common value is infinite.

Additional Proof Outline. The proof is obtained by progressively enlarging the class of functions for which the formula is known.

First, for a characteristic function of a measurable rectangle,

$$f = \chi_{A \times B},$$

we have

$$\int_{\Omega_1 \times \Omega_2} \chi_{A \times B} \, d(\mu_1 \times \mu_2) = \mu_1(A)\mu_2(B).$$

On the other hand,

$$\int_{\Omega_2} \chi_{A \times B}(x, y) \, d\mu_2(y) = \chi_A(x)\mu_2(B),$$

so

$$\int_{\Omega_1} \left(\int_{\Omega_2} \chi_{A \times B}(x, y) \, d\mu_2(y) \right) d\mu_1(x) = \mu_1(A)\mu_2(B).$$

By linearity, the formula extends to non-negative simple functions. Finally, if $f \geq 0$ is measurable, choose simple functions

$$0 \leq \varphi_n \uparrow f.$$

Apply the formula to φ_n and then use the Monotone Convergence Theorem on both the product integral and the iterated integrals. This yields Tonelli's theorem. □

Theorem 16.6 (Fubini's Theorem (Integrable Functions))**.** *If f is integrable on the product space (i.e.,*

$$\int_\Omega |f|\, d\mu < \infty,$$

which can be checked using Tonelli on $|f|$), then:

1. *For a.e. x, the slice*
$$f(x,\cdot)$$
is integrable on Ω_2.
2. *For a.e. y, the slice*
$$f(\cdot,y)$$
is integrable on Ω_1.
3. *The functions*
$$x \mapsto \int_{\Omega_2} f(x,y)\, d\mu_2(y)$$
and
$$y \mapsto \int_{\Omega_1} f(x,y)\, d\mu_1(x)$$
are integrable.
4. $$\int_\Omega f\, d\mu = \int_{\Omega_1} \left(\int_{\Omega_2} f\, d\mu_2 \right) d\mu_1 = \int_{\Omega_2} \left(\int_{\Omega_1} f\, d\mu_1 \right) d\mu_2.$$

Additional Proof. Since

$$f \in L^1(\Omega_1 \times \Omega_2),$$

we have

$$|f| \geq 0$$

and

$$\int_{\Omega_1 \times \Omega_2} |f|\, d(\mu_1 \times \mu_2) < \infty.$$

Applying Tonelli's theorem to $|f|$ gives

$$\int_{\Omega_1} \left(\int_{\Omega_2} |f(x,y)|\, d\mu_2(y) \right) d\mu_1(x) < \infty.$$

Therefore,

$$\int_{\Omega_2} |f(x,y)|\, d\mu_2(y) < \infty$$

for almost every x. Thus

$$f(x,\cdot) \in L^1(\Omega_2)$$

for almost every x.

Similarly,

$$f(\cdot,y) \in L^1(\Omega_1)$$

for almost every y.

Now write

$$f = f^+ - f^-.$$

Both f^+ and f^- are non-negative and integrable. Apply Tonelli to each:

$$\int f^+ = \int_{\Omega_1} \int_{\Omega_2} f^+, \qquad \int f^- = \int_{\Omega_1} \int_{\Omega_2} f^-.$$

Subtracting gives

$$\int f = \int_{\Omega_1} \left(\int_{\Omega_2} f \right),$$

and similarly for the reversed order. □

Remark 16.7. The practical distinction is:

$$\boxed{f \geq 0 \Longrightarrow \text{Tonelli}}$$

and

$$\boxed{\int |f| < \infty \Longrightarrow \text{Fubini.}}$$

For non-negative functions, the order can be exchanged even when the answer is $+\infty$. For signed functions, absolute integrability is the condition that guarantees the two iterated integrals and the product integral all agree.

16.4 Applications and Counterexamples

Example 16.8 (Computing Gaussian Integral)**.** To compute

$$I = \int_{-\infty}^{\infty} e^{-x^2}\, dx,$$

consider

$$I^2 = \int_{\mathbb{R}^2} e^{-(x^2+y^2)}\, dx\, dy.$$

Using Tonelli (function is positive), switch to polar coordinates (using change of variables in $\mathbb{R}^n$, a consequence of product measure theory):

$$\begin{aligned} I^2 &= \int_0^{2\pi} \int_0^{\infty} e^{-r^2} r\, dr\, d\theta \\ &= 2\pi \cdot \frac{1}{2} \\ &= \pi. \end{aligned}$$

Therefore,

$$I = \sqrt{\pi}.$$

Remark 16.9. Tonelli is applicable before we know whether the Gaussian integral is finite, because

$$e^{-(x^2+y^2)} \geq 0.$$

This is one reason Tonelli is particularly useful for evaluating positive multiple integrals.

Example 16.10 (Failure of Fubini for non-integrable functions)**.** Consider

$$f(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2}$$

on

$$(0,1) \times (0,1).$$

First integrate with respect to y:

$$\int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2}\, dy = \left[\frac{y}{x^2 + y^2}\right]_{y=0}^{y=1} = \frac{1}{1 + x^2}.$$

Hence

$$\int_0^1 \left(\int_0^1 \frac{x^2 - y^2}{(x^2 + y^2)^2}\, dy \right) dx = \int_0^1 \frac{1}{1 + x^2}\, dx = \frac{\pi}{4}.$$

By symmetry (swapping x, y flips the sign of f), the other iterated integral is

$$-\frac{\pi}{4}.$$

Since

$$\frac{\pi}{4} \neq -\frac{\pi}{4},$$

f is not absolutely integrable:

$$f \notin L^1.$$

Indeed,

$$\int |f| = \infty.$$

Remark 16.11. This example shows why the hypothesis

$$\int |f| < \infty$$

in Fubini's theorem cannot simply be deleted. Without absolute integrability,

$$\int \left(\int f(x, y)\, dy \right) dx$$

and

$$\int \left(\int f(x, y)\, dx \right) dy$$

may both exist but give different answers.

Remark 16.12 (Failure of σ-finiteness). Let

$$\Omega_1 = \Omega_2 = [0, 1].$$

Let μ_1 be Lebesgue measure and μ_2 be Counting measure. Let

$$D = \{(x, x) : x \in [0, 1]\}$$

be the diagonal.

Then

$$\int_{[0,1]} \left(\int_{[0,1]} \chi_D(x, y)\, d\mu_2(y) \right) d\mu_1(x) = \int_0^1 1\, dx = 1.$$

Indeed, for each fixed x, the vertical slice

$$\{y : (x, y) \in D\} = \{x\}$$

contains one point, so its counting measure is 1.

On the other hand,

$$\int_{[0,1]} \left(\int_{[0,1]} \chi_D(x,y)\, d\mu_1(x) \right) d\mu_2(y) = 0.$$

For each fixed y, the horizontal slice

$$\{x : (x,y) \in D\} = \{y\}$$

has Lebesgue measure zero.

Thus the two iterated integrals are different. Fubini fails because counting measure on $[0,1]$ is not σ-finite.

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